

11.1 Introduction

As is well known, the concept of the atom appears for the first time as a hypothesis about the structure of matter by Democritus (*c.* 460 BC–*c.* 370 BC). Democritus also proposed a rudimentary mechanics: Atoms moved in straight lines and through their collisions bodies were formed and destroyed. It is also interesting to recall that later Epicurus (341 BC–270 BC) and Lucretius (*c.* 94 BC–*c.* 49 BC) introduced a random, reasonless, deviation (*clinamen*) from the straight line in order to account for the contingency of our world. Though atomism remained a speculative theory for more than 2000 years, in modern ages it was still alive among many scientists and philosophers (Newton, Locke, Spinoza, etc.).

At the beginning of the nineteenth century John Dalton (1766–1844) introduced the concept of the atom in a scientific framework in order to explain some chemical phenomena. He was one of the earliest scientists to work on the structure of matter. He supposed that atoms were the smallest units of an element that enter into chemical combinations, and that a chemical element was composed entirely of one type of atom. Then, compounds contain atoms of two or more different elements, where the relative number of atoms of each element in a particular compound is always the same. Experimentally, he observed that, for each gram of hydrogen found in water there were always 8 grams of oxygen. Thus, he concluded that, in every pure substance, the same elements are in the same proportions. This observation is known as the law of constant composition. Moreover, it was possible to assign a mass to each given element in such a way that the ratio of different atoms in an element was a rational number (e.g. in water for each atom of oxygen there were two atoms of hydrogen). However, the most important of Dalton's assumptions was that atoms do not change their identities (neither the number nor the type of atoms change) in chemical reactions, which simply change the way the atoms are joined together. As a consequence, he also showed that there was no change in mass during a chemical reaction. This is called the law of conservation of mass.

A further step in transforming the atomic theory into a scientific hypothesis was the introduction in 1869 by D. I. Mendeleev (1834–1907) of the table of the elements that was later completed by himself. The table showed considerable predictive power in anticipating the discovery of some chemicals. At first, the table was taken as a numerical extravagance. The discovery of new elements that filled the gaps in the table forced scientists to take more seriously the hypothesis of atoms as elementary particles, whose combination allowed for the existence of the different elements in a specific way.

However, atoms only began to be considered to be real after the discovery of Brownian motion, the disordered motion of small particles suspended in a fluid or gaseous solution.¹ It is well known that Albert Einstein (1905) succeeded in formalizing the problem. Once the experimental measurements of Jean Perrin (1908) confirmed the theory (his results also led to a better estimation of Avogadro's number) and Robert Millikan (1910) succeeded in measuring the elementary electric charge, as well as the first trajectories of elementary particles were observed in the Wilson chamber (1912), atomic theory began to be an integral part of modern physics.

The first hypothesis about the structure of the atom was formulated in 1902 by J.J. Thompson, who suggested that the neutral atom was formed of a positively charged homogeneous sphere, within which the electrons were suspended in such a number and such a way as to give rise to a zero total charge. In 1911, Geiger and Marsden performed an experiment involving the scattering of alpha particles (see the end of Sec. 4.3) by a thin metal film that falsified Thompson's model by showing that most of the alpha particles passed through the film undeflected, whereas the few that were scattered underwent large deflection. If the positive charge (and the mass) of the atom were distributed uniformly within its volume, we would observe a large number of deflections at small angles. Therefore, Ernest Rutherford inferred that the positive charge of the atom and nearly all of its mass was in a tiny region, the nucleus, whose radius we now know to be of the order of 10^{-15} m, while the radius of the entire atom was known to be the order of 10^{-10} m. It was the strong electrostatic repulsion between the nucleus and the alpha particles that caused the large deflections. According to Rutherford's model the number of electrons was equal to the atomic number Z : they orbited circularly around the nucleus. This model presented two main problems: first, since an accelerating electric charge, according to Larmor's law,² emits electromagnetic radiation, the electrons should progressively lose energy and fall, with a spiral trajectory, into the nucleus. The second problem was related to its inability to explain the existence of line spectra (see Subsec. 1.5.4).

A further step toward the quantum-mechanical model of the atom was made in 1913 when Bohr introduced the hypothesis of the quantization of atomic levels, extending and generalizing Planck's quantization rule to the hydrogen atom (see also Subsec. 1.5.1). We know now that Bohr's result only represents a partial solution of the problem,³ whose completion can be obtained from the quantum-mechanical approach to the dynamics of the electrons. We shall see in this chapter the use of the Schrödinger equation in modern atomic theory.

After having introduced the general problem of the motion of quantum systems in the presence of a central potential, we shall introduce the theory of the hydrogen atom (in Sec. 11.2). In Sec. 11.3 we shall examine the problem of the atom in a magnetic field, while in Sec. 11.4 we shall introduce some relativistic corrections. However, in the present book we do not consider full relativistic effects as such and limit our exposition to approximation

¹ See [Parisi 2005a].

² See [Jackson 1962].

³ The reason why Bohr's model has to be considered an *ad hoc* hypothesis lies in the fact that there is no explanation concerning the stability of the discrete stationary electronic levels.

methods. In Sec. 11.5 we shall make use of different approximation methods for treating the theory of the helium atom. Finally, in Sec. 11.6 we shall consider the Thomas–Fermi and Hartree–Fock methods for dealing with the multi-electron problem.

11.2 Quantum theory of the hydrogen atom

Generally speaking, an atomic system is represented by a heavy central nucleus made by Z protons and a certain number of neutrons (see Box 7.1). Z is a positive integer number and is called the *atomic number*. Indicating with N the number of neutrons, the sum

$$A = Z + N \quad (11.1)$$

is called *mass number*, because, as we shall see, the mass of the neutron is approximately equal to the mass of the proton, which in turn is almost 2000 times the mass of the electron (see also Subsec. 6.3.1). On the other hand, the charge of the proton is, apart from the sign, approximately equal to that of the electron. Combining these facts, we may conclude that the atom is an analogue of the solar system with the nucleus replacing the Sun and the planets represented by the electrons. There is, however, a certain number of differences that need to be emphasized:

- The atom is a microscopic system, and therefore quantum effects not only cannot be neglected but even determine the nature of the system. As a consequence, electrons do not follow classical trajectories.
- The force that binds the solar system is the gravitational interaction. In an atomic system gravitation is negligible, and the relevant interaction is represented by the Coulomb force.
- While in the solar system, different planets may have considerable differences of mass, the electrons' mass is the same.

In order for the atom to be electrically neutral, it must have the same number of electrons and neutrons. Otherwise, we have positive or negative ions.

In what follows we shall be mainly concerned with the hydrogen atom, since it undoubtedly the simplest atomic system, in that its nucleus is made of a proton (without neutrons) and there is only one electron. For this reason, it can be considered a prototype of the two-body Kepler problem. The helium atom is the next simplest problem, since it involves two electrons.

11.2.1 The quantum Kepler problem

The classical Kepler problem deals with the gravitational force among celestial bodies in the solar system.⁴ As is well known from classical mechanics, when the number of

⁴ See [Goldstein 1950, Ch. 3].

interacting bodies is equal to or larger than three, there is no close-form solution of the set of differential equations describing this physical situation. In order to solve such a problem, Kepler treated the bodies as point masses and reduced the problem to a set of equations describing one-to-one interactions. An equivalent treatment is also useful for dealing with the description of the hydrogen atom. On an abstract plane, let us first consider the case of two particles (say a and b) that interact with a conservative force, i.e. whose Hamiltonian is given by

$$\hat{H} = \frac{(\hat{\mathbf{p}}^a)^2}{2m_a} + \frac{(\hat{\mathbf{p}}^b)^2}{2m_b} + V(\hat{r}), \quad (11.2)$$

where $\hat{\mathbf{r}} = \hat{\mathbf{r}}^a - \hat{\mathbf{r}}^b$. Note that in this case V does not depend on the angular coordinates θ and ϕ . We may therefore apply the same considerations we have made in Subsec. 6.2.2 about central potentials. As in the classical case, we move to the center-of-mass reference frame and write

$$\hat{\mathbf{R}} = \frac{m_a \hat{\mathbf{r}}^a + m_b \hat{\mathbf{r}}^b}{M}, \quad (11.3)$$

where $M = m_a + m_b$. We now show that, given the expressions

$$\hat{\mathbf{P}} = M\hat{\mathbf{R}} \quad \text{and} \quad \hat{\mathbf{p}} = m\hat{\mathbf{r}}, \quad (11.4)$$

where

$$m = \frac{m_a m_b}{M} \quad (11.5)$$

is the system reduced mass, the Hamiltonian of the system can be written as

$$\hat{H} = \frac{\hat{\mathbf{P}}^2}{2M} + \frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{r}) = \hat{H}_0 + \hat{H}_1, \quad (11.6)$$

where $\hat{H}_0 = \hat{\mathbf{P}}^2/2M$. Since

$$\hat{\mathbf{P}} = m_a \hat{\mathbf{r}}^a + m_b \hat{\mathbf{r}}^b = \hat{\mathbf{p}}^a + \hat{\mathbf{p}}^b, \quad (11.7)$$

we have

$$\frac{\hat{\mathbf{P}}^2}{2M} + \frac{\hat{\mathbf{p}}^2}{2m} = \frac{1}{2M} \left[(\hat{\mathbf{p}}^a)^2 + (\hat{\mathbf{p}}^b)^2 + 2\hat{\mathbf{p}}^a \hat{\mathbf{p}}^b \right] + \frac{m}{2} \left[(\hat{\mathbf{r}}^a)^2 + (\hat{\mathbf{r}}^b)^2 - 2\hat{\mathbf{r}}^a \hat{\mathbf{r}}^b \right], \quad (11.8)$$

from which the result (11.6) follows (see Prob. 11.1).

It is possible to prove that $\hat{\mathbf{P}}, \hat{\mathbf{R}}$ and $\hat{\mathbf{p}}, \hat{\mathbf{r}}$ are two pairs of conjugate variables classically (see Prob. 11.2) and quantum-mechanically, and the variables of the first pair commute with those of the second pair. For the quantum case, it is clear that $[\hat{r}_j^a, \hat{p}_k^a] = i\hbar\delta_{jk}$, where $\hat{r}_j^a$ and $\hat{p}_k^a$ are the j -th component of position and the k -th component of momentum of the first particle, respectively, and the same relations hold for the second particle. Moreover,

$$\begin{aligned} [\hat{R}_j, \hat{P}_k] &= \left[\frac{m_a}{M} \hat{r}_j^a + \frac{m_b}{M} \hat{r}_j^b, \hat{p}_k^a + \hat{p}_k^b \right] \\ &= \left(\frac{m_a}{M} + \frac{m_b}{M} \right) i\hbar\delta_{jk} = i\hbar\delta_{jk}, \end{aligned} \quad (11.9)$$

which is the desired result for the first pair of observables. For the second pair, we have

$$\begin{aligned} [\hat{r}_j, \hat{p}_k] &= \left[\hat{r}_j^a - \hat{r}_j^b, \frac{m}{m_a} \hat{p}_k^a - \frac{m}{m_b} \hat{p}_k^b \right] \\ &= \left(\frac{m}{m_a} + \frac{m}{m_b} \right) i\hbar \delta_{jk} = i\hbar \delta_{jk}, \end{aligned} \quad (11.10)$$

which is again the result we wished to have. In a similar way, one can show that the observables of the first pair commute with the observables of the second pair, i.e. (see Prob. 11.3)

$$[\hat{R}_j, \hat{p}_k] = [\hat{r}_j, \hat{P}_k] = [\hat{r}_j, \hat{R}_k] = [\hat{p}_j, \hat{P}_k] = 0. \quad (11.11)$$

We have divided the Hamiltonian into two parts (Eq. (11.6)), so that, if the total wave function is of the form

$$\psi(\mathbf{r}^a, \mathbf{r}^b) = \Phi(\mathbf{R}) \cdot \varphi(\mathbf{r}), \quad (11.12)$$

and is an eigenfunction of the total Hamiltonian, we have

$$\begin{aligned} \hat{H} \psi(\mathbf{r}^a, \mathbf{r}^b) &= (\hat{H}_0 + \hat{H}_1) \Phi(\mathbf{R}) \varphi(\mathbf{r}) \\ &= (E_R + E_r) \Phi(\mathbf{R}) \varphi(\mathbf{r}) = E \psi(\mathbf{r}^a, \mathbf{r}^b), \end{aligned} \quad (11.13a)$$

with

$$\hat{H}_0 \Phi(\mathbf{R}) = E_R \Phi(\mathbf{R}), \quad (11.13b)$$

$$\hat{H}_1 \varphi(\mathbf{r}) = E_r \varphi(\mathbf{r}). \quad (11.13c)$$

Equation (11.13b) represents the trivial eigenvalue equation for a free particle, and we will focus our attention on Eq. (11.13c), which, with the substitution made in Eq. (6.85), i.e.

$$\varphi(\mathbf{r}) = R(r) Y_{lm}(\phi, \theta), \quad (11.14)$$

can also be written as (see Eq. (6.87) and Prob. 6.12)

$$\left[-\frac{\hbar^2}{2mr^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{\hbar^2 \hat{\mathbf{L}}^2}{2mr^2} + V(r) - E_r \right] R(r) = 0. \quad (11.15)$$

Equation (11.14) is justified by the spherical symmetry of the problem, and the $Y_{lm}(\phi, \theta)$ are the spherical harmonics (6.65) (see Fig. 11.1).

11.2.2 The radial Schrödinger equation

When we deal with hydrogenoid atoms,⁵ the potential energy has the form

$$V(r) = -\frac{Ze^2}{r}, \quad (11.16)$$

⁵ Examples of hydrogenoid atoms are He^+ , Li^{++} , Be^{+++} , ...

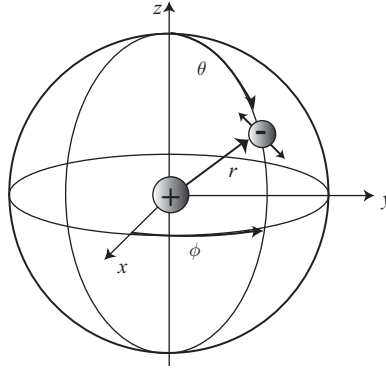


Figure 11.1

Electron coordinates in the atomic system. A negative electron is in motion around a positively charged nucleus. Even though it is not possible to define a quantum-mechanical trajectory of the electron, its position is individuated by the three spherical coordinates r, ϕ, θ : r represents the distance from the nucleus (see Fig. 6.3), and, via the energy eigenvalue, is associated with the *principal quantum number* n (see Fig. 6.9), ϕ is the azimuthal angle, and θ the polar angle (see Subsec. 6.1.3). Each (spin-1/2) electron may have a projection of the spin along the z -axis equal to $+\hbar/2$ or $-\hbar/2$ (see Sec. 6.3).

which represents the electrostatic interaction between a nucleus with charge Ze and only one electron with charge $-e$, where

$$e = 1.602\,176\,53 \times 10^{-19} \text{C}, \quad (11.17)$$

C being the Coulomb charge unit. Then, Eq. (11.15) (see Eq. (6.92)) can be rewritten as

$$\xi'' + \frac{2m}{\hbar^2} \left[E_r - \frac{\hbar^2}{2m} \frac{l(l+1)}{r^2} + \frac{Ze^2}{r} \right] \xi = 0, \quad (11.18)$$

where

$$R(r) = \frac{\xi(r)}{r}. \quad (11.19)$$

Figure 11.2 shows that the total potential resulting from the sum of the attractive Coulomb potential and the repulsive centrifugal term has a well shape (see Prob. 11.4).

Before looking for the solution of Eq. (11.18), we wish to introduce a general remark. We have here three natural constants, $\hbar$, e , and m , where

$$\frac{1}{m} = \frac{1}{m_e} + \frac{1}{m_n}, \quad (11.20)$$

and m_e is the electron mass; m_n is the nucleus mass, which is approximately equal to $Z m_p$; and m_p is the mass of the proton, which is about $1840 m_e$.⁶ Therefore, to an excellent

⁶ The mass of a “free” proton and a “free” neutron is $1.672\,621\,58 \times 10^{-27} \text{ kg}$ and $1.674\,927\,16 \times 10^{-27} \text{ kg}$, respectively. The sum amounts to $3.347\,548\,74 \times 10^{-27} \text{ kg}$. However, the mass of a “free” deuteron is only $3.343\,583\,09 \times 10^{-27} \text{ kg}$. The difference is $-0.003\,965\,65 \times 10^{-27} \text{ kg}$. This problem is called “mass defect.” The reason is that the binding of the proton and neutron to form the deuteron releases a certain amount of energy which is equivalent to the mass defect.

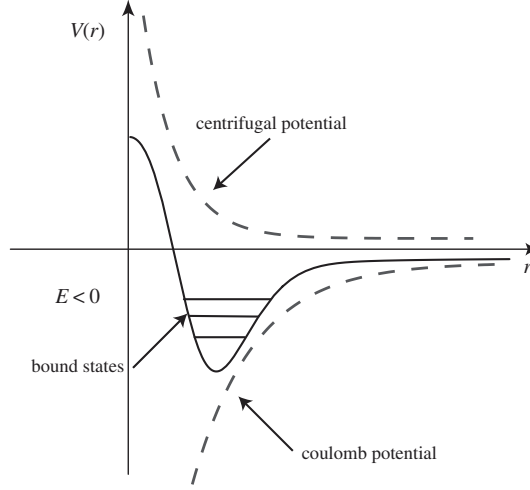


Figure 11.2

Resulting potential in the hydrogen atom (see also Fig. 4.9). The sum of the centrifugal potential $\frac{\hbar^2 l(l+1)}{2mr^2}$ and of the Coulomb potential $-Ze^2/r$ gives rise to the final well-shaped resulting potential. The negative-energy bound states are also schematically shown.

approximation (see Prob. 11.5), the reduced mass m is equal to the electron mass. With these constants we can build the quantities

$$r_0 = \frac{\hbar^2}{me^2} = 0.529\,177\,208\,59(36) \times 10^{-10} \text{ m}, \quad E_0 = \frac{me^4}{\hbar^2} = 27.2 \text{ eV}, \quad (11.21)$$

where r_0 is called *Bohr's radius*, and has an uncertainty of $0.000\,000\,000\,36 \times 10^{-10} \text{ m}$. It is interesting to note that with the same constants it is also possible to define Bohr's magneton or the characteristic unity of magnetic momenta (see also Eq. (6.167)), i.e.

$$\mu_B = \frac{e\hbar}{2mc}. \quad (11.22)$$

Let us now come back to the Schrödinger equation (11.18) in the case where $Z = 1$ (hydrogen atom). It is possible to write (see also Prob. 11.6)

$$r_0^2 \frac{d^2}{dr^2} \xi = \frac{d^2}{d\tilde{r}^2} \xi, \quad (11.23a)$$

$$\left[\frac{d^2}{d\tilde{r}^2} + \frac{2}{\tilde{r}} - \frac{l(l+1)}{\tilde{r}^2} + 2\tilde{E} \right] \xi(\tilde{r}) = 0, \quad (11.23b)$$

where

$$\tilde{r} = \frac{r}{r_0} \quad \text{and} \quad \tilde{E} = \frac{E}{E_0}. \quad (11.24)$$

Consider the change of variable

$$n^2 = -\frac{1}{2\tilde{E}}. \quad (11.25)$$

Box 11.1

Confluent hypergeometric functions

A confluent hypergeometric function has the general form [Gradstein/Ryshik 1981, 9.210, 9.216]

$$F(\alpha; \gamma; z) = 1 + \frac{\alpha}{\gamma} \frac{z}{1!} + \frac{\alpha(\alpha+1)}{\gamma(\gamma+1)} \frac{z^2}{2!} + \frac{\alpha(\alpha+1)(\alpha+2)}{\gamma(\gamma+1)(\gamma+2)} \frac{z^3}{3!} + \dots, \quad (11.26)$$

where z is the (complex) variable and α and γ are two complex parameters. $F(\alpha; \gamma; z)$ is the solution of the differential equation

$$z \frac{d^2}{dz^2} f(z) + (\gamma - z) \frac{d}{dz} f(z) - \alpha f(z) = 0, \quad (11.27)$$

and its asymptotic behavior for $|z| \rightarrow \infty$ is given by

$$F(\alpha; \gamma; z) \rightarrow e^{-i\pi\alpha} \frac{\Gamma(\gamma)}{\Gamma(\gamma-\alpha)} z^{-\alpha} + \frac{\Gamma(\gamma)}{\Gamma(\alpha)} e^z z^{\alpha-\gamma}, \quad (11.28)$$

where Γ is the Euler gamma function [Gradstein/Ryshik 1981, 8.3]. Equation (11.28) is not valid for $\alpha = -n$, with $n = 0, 1, 2, \dots$, in which case F becomes a polynomial of degree n in z . An important property of F is represented by *Kummer's transformation* [Gradstein/Ryshik 1981, 9.212]

$$F(\alpha; \gamma; z) = e^z F(\gamma - \alpha; \gamma; -z). \quad (11.29)$$

The asymptotic behavior of the solution of Eq. (11.23b) for $\tilde{r} \rightarrow \infty$ and for $\tilde{r} \rightarrow 0$ suggests the ansatz⁷

$$\xi(\tilde{r}) = \tilde{r}^{l+1} e^{-\frac{\tilde{r}}{n}} W(\tilde{r}). \quad (11.30)$$

Let us use the relations

$$\frac{\partial}{\partial \tilde{r}} \xi(\tilde{r}) = \left(\frac{l+1}{\tilde{r}} - \frac{1}{n} + \frac{W'}{W} \right) \tilde{r}^{l+1} e^{-\frac{\tilde{r}}{n}} W, \quad (11.31a)$$

$$\frac{\partial^2}{\partial \tilde{r}^2} \xi(\tilde{r}) = \left[-\frac{l+1}{\tilde{r}^2} + \frac{W''}{W} - \frac{(W')^2}{W^2} + \left(\frac{l+1}{\tilde{r}} - \frac{1}{n} + \frac{W'}{W} \right)^2 \right] \tilde{r}^{l+1} e^{-\frac{\tilde{r}}{n}} W, \quad (11.31b)$$

in order to derive (see Prob. 11.7)

$$\left[\tilde{r} W'' + 2 \left(1 + l - \frac{\tilde{r}}{n} \right) W' + 2 \frac{n-l-1}{n} W \right] \xi(\tilde{r}) = 0, \quad (11.32)$$

where the derivatives of $W(\tilde{r})$ are taken with respect to $\tilde{r}$. With the change of variable

$$\eta = \frac{2\tilde{r}}{|n|}, \quad (11.33)$$

⁷ Some textbooks incorrectly infer the form of the solution from the asymptotic behavior of Eq. (11.23b). We choose here to follow a more general approach (see [Loinger 2003]).

we may distinguish two cases:

- $n > 0$, i.e. $n = |n|$;
- $n < 0$, i.e. $n = -|n|$.

We shall show in the following that the two cases lead to the same result.

In the first case ($n = |n|$), we obtain

$$\eta W''(\eta) + [2(l+1) - \eta] W'(\eta) + (|n| - l - 1) W(\eta) = 0, \quad (11.34)$$

where the derivatives of $W(\eta)$ are now taken with respect to η . The solution of Eq. (11.34) is

$$W(\eta) = F(-|n| + l + 1; 2l + 2; \eta), \quad (11.35)$$

where $F(\alpha; \gamma; z)$ is the confluent hypergeometric function (see Box 11.1), that is, a function of the form

$$\begin{aligned} F(-|n| + l + 1; 2l + 2; \eta) = 1 + & \frac{(-n + l + 1) \eta}{(2l + 2) 1!} \\ & + \frac{(-n + l + 1)(-n + l + 1 + 1) \eta^2}{(2l + 2)(2l + 2 + 1) 2!} \\ & + \frac{(-n + l + 1)(-n + l + 1 + 1)[(-n + l + 1) + 2] \eta^3}{(2l + 2)(2l + 2 + 1)(2l + 2 + 2) 3!} \\ & + \dots \end{aligned} \quad (11.36)$$

This in turn justifies our ansatz. Then, in the limit $\tilde{r} \rightarrow \infty$ ($\eta \rightarrow \infty$), taking into account the expression (11.28), from Eq. (11.30) we have

$$\xi(\tilde{r}) = \tilde{r}^{l+1} e^{-\frac{\tilde{r}}{|n|}} F(-|n| + l + 1; 2l + 2; \eta) \rightarrow g(\tilde{r}) e^{\frac{\tilde{r}}{|n|}}, \quad (11.37)$$

where we recall substitution (11.33), and $g(\tilde{r}) \simeq \tilde{r}$.

In the second case ($n = -|n|$), we obtain the differential equation

$$\eta W'' + [2(l+1) + \eta] W' + (|n| + l + 1) W = 0, \quad (11.38)$$

whose solution is (see Prob. 11.8)

$$W(\eta) = F(|n| + l + 1; 2l + 2; -\eta), \quad (11.39)$$

where F is again the confluent hypergeometric function with the form

$$\begin{aligned} F(|n| + l + 1; 2l + 2; -\eta) = 1 + & \frac{(n + l + 1) - \eta}{(2l + 2) 1!} \\ & + \frac{(n + l + 1)(n + l + 1 + 1) (-\eta)^2}{(2l + 2)(2l + 2 + 1) 2!} \\ & + \frac{(n + l + 1)(n + l + 1 + 1)[(n + l + 1) + 2] (-\eta)^3}{(2l + 2)(2l + 2 + 1)(2l + 2 + 2) 3!} \\ & + \dots \end{aligned} \quad (11.40)$$

We make use now of Kummer's transformation

$$F(|n| + l + 1; 2l + 2; -\eta) = e^{-\eta} F(-|n| + l + 1; 2l + 2; \eta), \quad (11.41)$$

and, for $\tilde{r} \rightarrow \infty$ ($\eta \rightarrow \infty$), we obtain

$$\begin{aligned}\xi(\tilde{r}) &= \tilde{r}^{l+1} e^{\frac{\tilde{r}}{|n|}} F(|n| + l + 1; 2l + 2; -\eta) \\ &= \tilde{r}^{l+1} e^{\frac{\tilde{r}}{|n|}} e^{-\frac{2\tilde{r}}{|n|}} F(-|n| + l + 1; 2l + 2; \eta) \rightarrow g(\tilde{r}) e^{\frac{\tilde{r}}{|n|}}.\end{aligned}\quad (11.42)$$

Equation (11.42), as expected, proves that the two cases ($n < 0$ and $n > 0$) lead to the same result. Since there is dependence on the sign of E (negative for bound states), but not on the sign of n , from now on we may take $n > 0$. From the expansion (11.36) we learn that $W(\eta)$ bears the general form

$$W(\eta) = \sum_{j=0}^{\infty} c_j \eta^j, \quad (11.43)$$

where

$$c_{j+1} = \frac{j + l + 1 - n}{(j + 1)[j + 2(l + 1)]} c_j. \quad (11.44)$$

The first three terms are given by

$$c_0 = 1, \quad c_1 = \frac{(l + 1 - n)}{2(l + 1)}, \quad c_2 = \frac{(l + 2) - n}{2(2l + 3)} c_1. \quad (11.45)$$

11.2.3 Eigenvalues and eigenfunctions

Let us have a closer look at Eqs. (11.43) and (11.44). We have here two possibilities:

- One of the c_j , say c_k , is zero and consequently all other c_j with $j > k$ are also zero, and $W(\eta)$ is a polynomial.
- Otherwise, using the fact that

$$\frac{c_{j+1}}{c_j} \rightarrow \frac{1}{j}, \quad (11.46)$$

for $j \rightarrow \infty$, we would have $c_j \simeq 1/(j - 1)!$, and consequently also

$$W(\eta) = \sum_{j=0}^{\infty} c_j \eta^j \propto \eta e^{\eta}. \quad (11.47)$$

This second possibility implies that $W(\eta)$ is not a polynomial and grows as e^{η} for $\eta \rightarrow \infty$. This alternative must be discarded, and we must therefore require that the series (11.43) or (11.36) be truncated at a certain value $j = k$. In order for this to happen, it is necessary that

$$n = l + 1 + k, \quad (11.48)$$

where k must be a non-negative integer. In other words, the radial wave function contains only a finite number of terms. Since l is an integer number, also n must be an integer num-

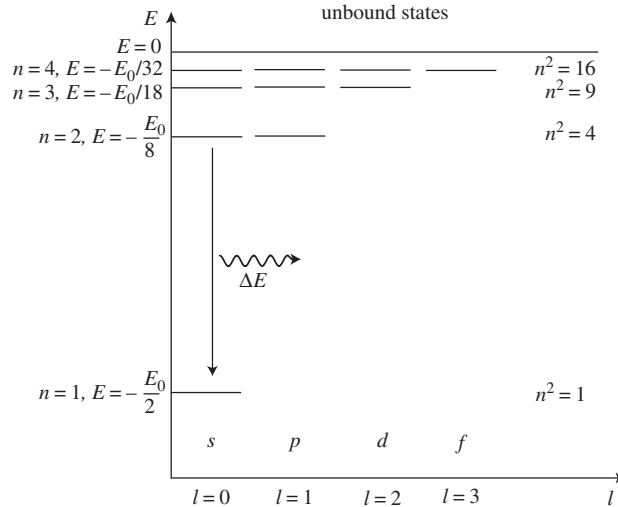


Figure 11.3

The Grotrian scheme: a schematic representation of the populated energy levels as a function of the two quantum numbers n and l . The arrow shows the transition from the $n = 2, l = 0$ state to the $n = 1, l = 0$ state, with emission of a photon of energy $\Delta E = 3E_0/8$.

ber. For each integer value of n , the corresponding value of energy, given the definitions (11.24), is then given by

$$E_n = E_0 \tilde{E} = -\frac{1}{2n^2} E_0, \quad (11.49)$$

where $n = 1, 2, \dots$. It is then clear that the energy levels of the bound states become more and more dense as n goes to infinity, i.e. as the energy approaches the limit $E = 0$ (see Fig. 11.3). The number n is also called the *principal quantum number*. The energy levels with the same value of l may be grouped together. In this way one may build the Lyman ($l = 0$), Balmer ($l = 1$), Paschen ($l = 2$), Blackett ($l = 3$), and Pfund ($l = 4$) series (see also the end of Subsec. 6.1.4). The number of eigenstates for each given value of n is equal to n^2 (see Prob. 11.9). As we have seen, the energy, for a given n , does not depend on l . This may look bizarre, as l appears in the Hamiltonian. Such an l -degeneracy is only accidental.⁸ As we shall see in the following, relativistic corrections may remove such a degeneracy.

The ground state $n = 1$ ($l = 0$) is obviously stable. However, transitions from upper levels to lower ones involve the emission of photons with energy equal to the difference between the energies of the these levels. For example, the transition from the $n = 2, l = 0$ state to the $n = 1, l = 0$ state involves the emission of a photon of energy:

$$\Delta E = -\frac{E_0}{8} - \left(-\frac{E_0}{2}\right) = \frac{3}{8} E_0. \quad (11.50)$$

⁸ It occurs only for the Coulomb potential proportional to $1/r$. For instance, a different degeneracy is present in the case of the three-dimensional harmonic oscillator (see Subsec. 6.2.4).

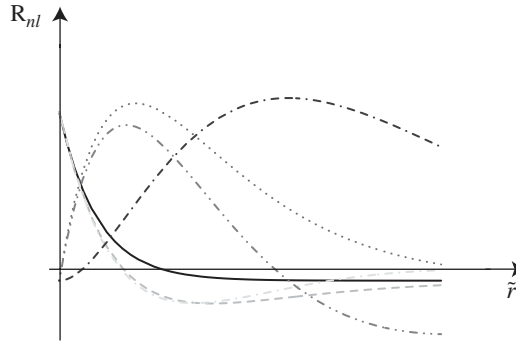


Figure 11.4

Schematic plot of the first few radial eigenfunctions of the hydrogenoid atom: R_{10} (solid line), R_{20} (dashed line), R_{21} (dotted line), R_{30} (dot-dashed line), R_{31} (dot-dot-dashed line), and R_{32} (dash-dot-dot line).

For completeness in the following we present explicit expressions for the first few functions R_{nl} (see Fig. 11.4 and Prob. 11.10):

$$R_{10}(\tilde{r}) \propto e^{-\tilde{r}}, \quad (11.51a)$$

$$R_{20}(\tilde{r}) \propto e^{-\frac{\tilde{r}}{2}} \left(1 - \frac{\tilde{r}}{2}\right), \quad (11.51b)$$

$$R_{21}(\tilde{r}) \propto \tilde{r} e^{-\frac{\tilde{r}}{2}}, \quad (11.51c)$$

$$R_{30}(\tilde{r}) \propto e^{-\frac{\tilde{r}}{3}} \left(1 - \eta + \frac{1}{6}\eta^2\right) = e^{-\frac{\tilde{r}}{3}} \left(1 - \frac{2}{3}\tilde{r} + \frac{2}{27}\tilde{r}^2\right), \quad (11.51d)$$

$$R_{31}(\tilde{r}) \propto \eta \left(1 - \frac{1}{4}\eta\right) e^{-\frac{\eta}{2}} = \frac{2}{3}\tilde{r} \left(1 - \frac{1}{6}\tilde{r}\right) e^{-\frac{\tilde{r}}{3}}, \quad (11.51e)$$

$$R_{32}(\tilde{r}) \propto \eta^2 e^{-\frac{\eta}{2}} = \frac{4}{9}\tilde{r}^2 e^{-\frac{\tilde{r}}{3}}. \quad (11.51f)$$

As expected (see property (ii) of Subsec. 3.2.3), R_{10} has no zeros, R_{20} has one zero (for $r = 2r_0$), R_{30} has two zeros, and so on. In order to find the correct normalization factors for the radial wave functions $R_{nl}(r)$, one has to impose the condition (see Prob. 11.11)

$$\int_0^{+\infty} dr |R_{nl}(r)|^2 r^2 = 1. \quad (11.52)$$

It is particularly interesting to study the radial probability density for the electron, since it may give an idea on at what distance from the nucleus the probability of finding the electron is maximal (see Fig. 11.5). Such a probability density (see Prob. 11.12) is given by

$$\wp_{nl}(\tilde{r}) = |R_{nl}(r)|^2 r^2. \quad (11.53)$$

11.2.4 Total eigenfunctions and hydrogenoid atoms

Recalling the definition of the total eigenfunction $\varphi(\mathbf{r})$ (Eq. (11.14)), we may now use the result for the radial wave function, which is given by Eqs. (11.19), (11.30), (11.43), and

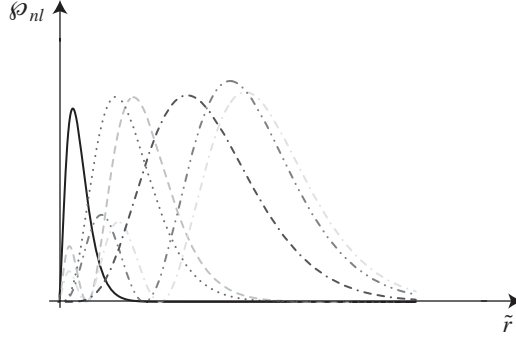


Figure 11.5

Schematic plot of the radial probability densities $\rho_{nl}(\tilde{r}) = |R_{nl}(\tilde{r})|^2 \tilde{r}^2$ for the first few radial eigenfunctions of the hydrogenoid atom (same convention as in Fig. 11.4).

(11.44), and Eq. (6.65) for the spherical harmonics in order to build the explicit form of

$$\varphi_{nlm}(r, \theta, \phi) = R_{nl}(r) Y_{lm}(\theta, \phi). \quad (11.54)$$

In the case of hydrogenoid atoms, the zero-angular momentum total eigenfunctions $\varphi_{nlm}(r, \theta, \phi)$ for $n = 1, 2, 3$, are (see Probs. 11.14 and 11.15)

$$\varphi_{100}(\mathbf{r}) = \frac{1}{\sqrt{\pi}} \left(\frac{Z}{r_0} \right)^{\frac{3}{2}} e^{-\frac{Zr}{r_0}}, \quad (11.55a)$$

$$\varphi_{200}(\mathbf{r}) = \frac{1}{4\sqrt{2\pi}} \left(\frac{Z}{r_0} \right)^{\frac{3}{2}} \left(2 - \frac{Zr}{r_0} \right) e^{-\frac{Zr}{2r_0}}, \quad (11.55b)$$

$$\varphi_{300}(\mathbf{r}) = \frac{1}{81\sqrt{3\pi}} \left(\frac{Z}{r_0} \right)^{\frac{3}{2}} \left(27 - 18\frac{Zr}{r_0} + \frac{2Z^2 r^2}{r_0^2} \right) e^{-\frac{Zr}{3r_0}}. \quad (11.55c)$$

11.3 Atom and magnetic field

As is natural to expect, the presence of a constant and homogenous magnetic field modifies the energy levels of an electron in an atomic system and may be considered as a perturbation. This effect is basically due the magnetic interaction between the external magnetic field and the intrinsic magnetic momentum of the electron. As we shall see, depending on the strength of the magnetic field, it is possible to distinguish between two regimes that correspond to two different effects: the Paschen–Bach effect (Subsec. 11.3.2) and the Zeeman effect (Subsec. 11.3.3).

Before entering into the detailed investigation of the effects produced by a magnetic field, it is necessary to consider another type of interaction that is always present in an atom (when $l \neq 0$) and goes under the name of the spin–orbit interaction. As we shall see in the next section, this interaction has a relativistic nature.

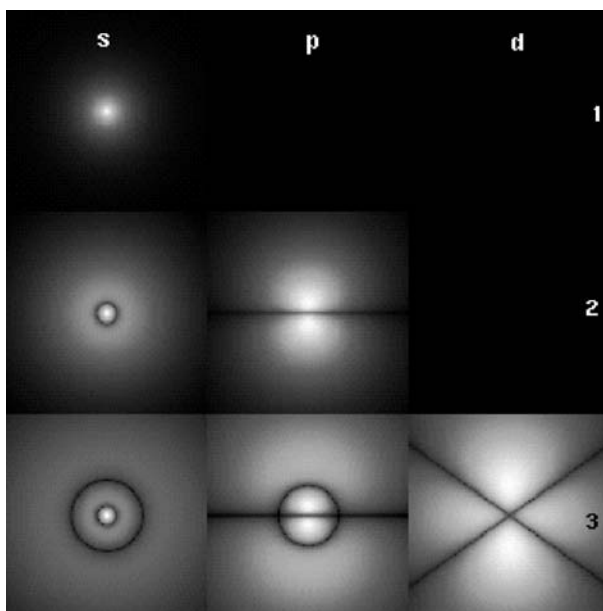


Figure 11.6

Pictorial representation (density plot) of the spatial probability density corresponding to the wave functions of an electron in a hydrogen atom possessing definite values of energy (increasing downward: $n = 1, 2, 3, \dots$) and of angular momentum (increasing across: $s, p, d, \dots$). Brighter areas correspond to higher probability density for a position measurement. Adapted from the web page www.physicsdaily.com/physics/Quantum_mechanics.

11.3.1 Spin-orbit interaction

We recall that Uhlenbeck and Goudsmit pointed out that certain spectroscopical lines, if observed with high precision (with a relative accuracy of the order of 10^{-5}), turn out to be doublets instead of single lines (see Fig. 6.9).⁹ They made the hypothesis that this was due to the presence of a magnetic momentum intrinsic to the electron, i.e. the spin (see Sec. 6.3).

Let us consider the reference frame where the electron is at rest. The proton's motion induces a current that generates a magnetic field which interacts with the electron. The electrostatic interaction between electron and proton is represented by the potential defined in Eq. (11.16), which corresponds to the electric field¹⁰

$$\mathbf{E} = \frac{Ze}{r^3} \mathbf{r}. \quad (11.56)$$

⁹ See [Uhlenbeck/Goudsmit 1925, Uhlenbeck/Goudsmit 1926].

¹⁰ We consider here electric and magnetic fields in unquantized form (and for this reason we have omitted the operatorial expression in the first few equations). The quantization of electromagnetic fields will be the subject of Ch. 13.

On the other hand, the “proton current” generates a magnetic field determined by (see Eq. (6.168))

$$\mathbf{B}_p = \mathbf{E} \times \frac{\mathbf{v}}{c} = \frac{Ze}{cr^3} \mathbf{r} \times \mathbf{v} = \frac{Ze}{m_e c} \frac{1}{r^3} \mathbf{L} = B_p \mathbf{k}, \quad (11.57)$$

where m_e is the electron mass, $\mathbf{L}$ its angular momentum, and $\mathbf{k}$ is the z -axis versor. Since the electron has an intrinsic magnetic momentum

$$\boldsymbol{\mu}_s = \frac{1}{2} \frac{e}{m_e c} \mathbf{S}, \quad (11.58)$$

there will be an interaction potential energy (see Subsec. 6.3.3) that in the nucleus reference frame¹¹ has the form

$$V = -\boldsymbol{\mu}_s \cdot \mathbf{B}_p = \frac{Ze^2}{2m_e^2 c^2 r^3} \mathbf{L} \cdot \mathbf{S} = \frac{1}{2} \frac{Z\alpha\hbar}{m_e^2 c} \frac{\mathbf{L} \cdot \mathbf{S}}{r^3}, \quad (11.59)$$

where

$$\alpha = \frac{e^2}{\hbar c} \simeq 137^{-1} \quad (11.60)$$

is the fine-structure constant. Due to the smallness of the fine-structure constant, such a spin–orbit interaction may be treated as a weak perturbation. However, this perturbation is sufficient to render l and m_l as “bad” quantum numbers. The new good quantum numbers are then j and m_j (see Sec. 6.4). In fact, in the quantized framework, from Eq. (6.188) we obtain

$$\hat{J}^2 = \hat{L}^2 + \hat{S}^2 + 2\hat{\mathbf{L}} \cdot \hat{\mathbf{S}}, \quad (11.61)$$

and

$$\hat{\mathbf{L}} \cdot \hat{\mathbf{S}} = \frac{1}{2} (\hat{J}^2 - \hat{L}^2 - \hat{S}^2). \quad (11.62)$$

In order to find the first-order perturbational correction to the energy level (see Eq. (10.12)), we have to compute the mean value

$$\langle \varphi_{nlm}^{(0)} | V | \varphi_{nlm}^{(0)} \rangle = \frac{Ze^2}{2m_e^2 c^2} \langle \hat{\mathbf{L}} \cdot \hat{\mathbf{S}} \rangle_{\text{ang,spin}} \left\langle \frac{1}{r^3} \right\rangle_{\text{rad}}, \quad (11.63)$$

where $\langle \cdot \rangle_{\text{ang,spin}}$ denotes the mean value on the angular and spin variables, and

$$\left\langle \frac{1}{r^3} \right\rangle_{\text{rad}} = \frac{Z^3}{r_0^3 n^3 l \left(l + \frac{1}{2}\right) (l+1)}. \quad (11.64)$$

In conclusion, the first-order correction to the energy level is given by

$$E_{nlm}^{(1)} = \frac{Z^4 \alpha e^6}{2c\hbar^5 n^3 l \left(l + \frac{1}{2}\right) (l+1)} \langle \hat{\mathbf{L}} \cdot \hat{\mathbf{S}} \rangle_{\text{ang,spin}} = \kappa \langle \hat{\mathbf{I}} \cdot \hat{\mathbf{S}} \rangle_{\text{ang,spin}}, \quad (11.65)$$

¹¹ See also [Jackson 1962, 546].

where

$$\kappa = \frac{Z^4 \alpha e^6 m_e}{2c \hbar^3 n^3 l \left(l + \frac{1}{2}\right) (l+1)} = \frac{Z^4 \alpha^4 m_e c^2}{2n^3 l \left(l + \frac{1}{2}\right) (l+1)}, \quad (11.66)$$

and

$$\langle \hat{\mathbf{l}} \cdot \hat{\mathbf{s}} \rangle_{\text{ang, spin}} = \frac{1}{2} [j(j+1) - l(l+1) - s(s+1)]. \quad (11.67)$$

Since $\hat{s} = 1/2$, we shall necessarily have $j = l \pm 1/2$ in Eq. (11.67). From Eq. (11.66) we expect the spin–orbit interaction to decrease as $1/n^3$ and $1/l^3$, so that the doublets reduce their separations as n and l increase (see Fig. 6.9 and Prob. 11.16).

Let us now go back to the problem of the behavior of an atom in an external homogeneous magnetic field $\mathbf{B}_{\text{ext}}$. Taking into account both the magnetic momenta induced by the orbital angular momentum and spin (see Eqs. (6.166) and (6.169)), and the spin–orbit interaction (Eq. (11.59)), the total Hamiltonian will be given by

$$\hat{H} = \hat{H}_A - \frac{e\hbar}{2m_e c} (\hat{\mathbf{l}} + 2\hat{\mathbf{s}}) \cdot \mathbf{B}_{\text{ext}} - f(r) \hat{\mathbf{l}} \cdot \hat{\mathbf{s}}, \quad (11.68)$$

where $\hat{H}_A$ is the ordinary atomic Hamiltonian (11.2), and

$$f(r) = \frac{Ze^2 \hbar^2}{2m_e^2 c^2 r^3}. \quad (11.69)$$

It should be noted that

$$\kappa = \langle f(r) \rangle_{\text{rad}}. \quad (11.70)$$

To find the solution of the general problem with Hamiltonian (11.68) is impossible. However, a perturbational approach (see Sec. 10.1) is viable in specific situations. In particular, Eq. (11.68) shows that we are in the presence of two perturbational terms in the Hamiltonian. We have necessarily to ask which of the two is the more relevant. If the magnetic term due to the external field is much larger than the spin–orbit interaction, the coupling between the orbital magnetic momentum and the spin magnetic momentum is negligible. In this case, $\hat{\mathbf{l}}$ and $\hat{\mathbf{s}}$ turn out to be decoupled, and m_l and m_s are again “good” quantum numbers. We denote this situation as the *Paschen–Bach effect*. In the opposite case, $\hat{\mathbf{l}}$ and $\hat{\mathbf{s}}$ are coupled, and the good quantum numbers are represented by j and m_j , giving rise to the *Zeeman effect*.

It should be noted that this distinction, given the strength B_{ext} of the magnetic field, depends on the excitation level of the atom: increasing n , the spin–orbit interaction decreases, and the highly excited levels (large n) feel the Paschen–Bach effect, whereas levels with small n feel the Zeeman effect.

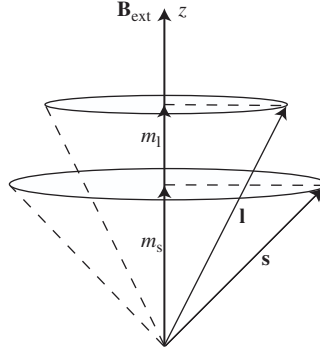


Figure 11.7

Landé semiclassical vectorial model for the Paschen-Bach effect. In order to compute the contribution of the spin-orbit correction, the spin and orbital angular momentum vectors can be thought of as precessing about the external magnetic field directed along the z -axis. As a consequence, only the time averaged components of $\hat{\mathbf{l}}$ and $\hat{\mathbf{s}}$ (directed along the z -axis) contribute to the scalar product in the third term of Eq. (11.71).

11.3.2 Paschen-Bach effect

Since the third term in Eq. (11.68) is negligible with respect to the second, we are allowed to separate the Hamiltonian according to the Dirac picture

$$\hat{H} = \hat{H}_A - \frac{e\hbar}{2m_e c} B_{\text{ext}} (\hat{l}_z + 2\hat{s}_z) - f(r) \hat{\mathbf{l}} \cdot \hat{\mathbf{s}} = \hat{H}_0 + \hat{H}_I, \quad (11.71)$$

where

$$\hat{H}_0 = \hat{H}_A - \frac{e\hbar}{2m_e c} B_{\text{ext}} (\hat{l}_z + 2\hat{s}_z), \quad (11.72)$$

and

$$\hat{H}_I = -f(r) \hat{\mathbf{l}} \cdot \hat{\mathbf{s}}. \quad (11.73)$$

We have assumed that the external magnetic field is along the z direction, i.e.

$$\mathbf{B}_{\text{ext}} = B_{\text{ext}} \mathbf{k}. \quad (11.74)$$

The eigenfunctions of $\hat{H}_A$ are also eigenfunctions of $\hat{l}_z$ and s_z . As a consequence, the solution of the eigenvalue equation for $\hat{H}_0$ is straightforward, i.e.

$$\begin{aligned} \hat{H}_0 \varphi_{nlm_l m_s} &= \left[\hat{H}_A - \frac{e\hbar}{2m_e c} B_{\text{ext}} (\hat{l}_z + 2\hat{s}_z) \right] \varphi_{nlm_l m_s} \\ &= \left[E_{nl}^{(0)} - \mu_B B_{\text{ext}} (m_l + 2m_s) \right] \varphi_{nlm_l m_s}, \end{aligned} \quad (11.75)$$

where $E_{nl}^{(0)}$ is the unperturbed atomic energy level. We have now to add the spin-orbit perturbation. To this aim, we compute the first-order perturbational correction – given by $\hat{H}_I$ – to the energy level, i.e. we have to calculate the mean value of $-f(r) \hat{\mathbf{l}} \cdot \hat{\mathbf{s}}$ on the unperturbed states of the atom. In order to perform this calculation, we take advantage of the so-called (semiclassical) Landé's *vectorial model* (see Figs. 6.12 and 11.7): the spin

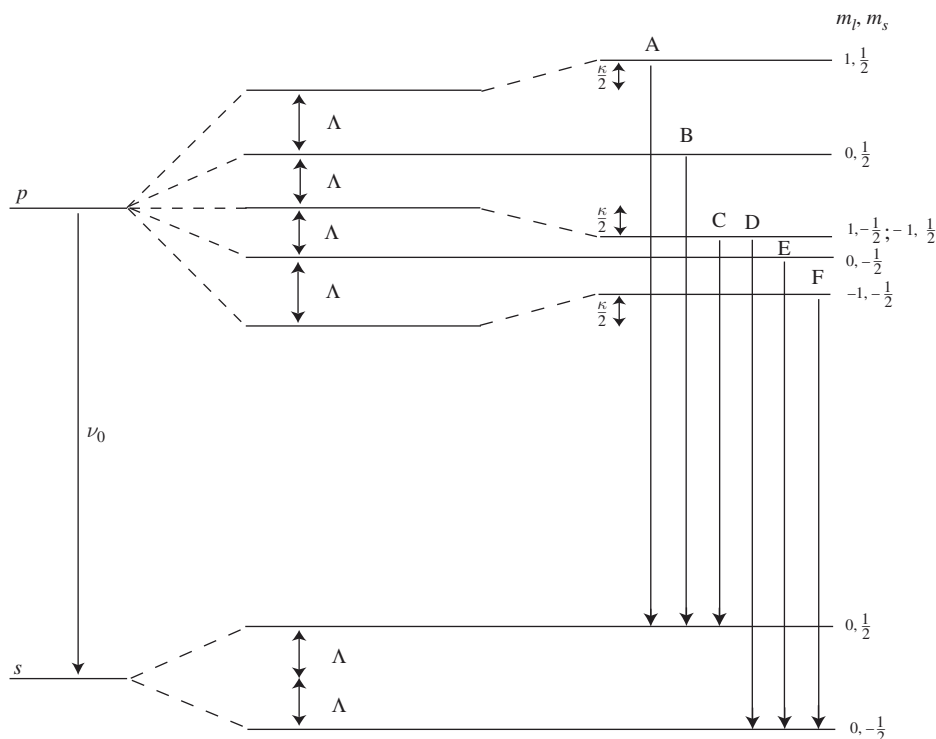


Figure 11.8

Diagram of the energy corrections to the s and p levels due to the Paschen–Bach effect. In the case of the s level, $l = 0$ and therefore there is no spin–orbit correction. However, the correction due to the external magnetic field gives rise to the doublet $(0, 1/2)$, $(0, -1/2)$, where the first number is m_l and the second one is m_s . The two components of the doublet are here separated by 2Λ . As for the p level ($m_l = -1, 0, 1$, $m_s = -1/2, 1/2$), there are five possibilities given by the five possible results of $m_l + 2m_s$. Apart from the $(0, 1/2)$ and $(0, -1/2)$ sublevels, for which the spin–orbit correction $\kappa \hat{l} \cdot \hat{s}$ is obviously zero, the sublevels are further corrected by the spin–orbit perturbation. The levels $(1, -1/2)$ and $(-1, 1/2)$ are degenerate, since both the magnetic and the spin–orbit corrections contribute exactly to the same extent.

and orbital angular momentum vectors can be thought of as precessing about the external magnetic field vector. Then, only $\hat{l}_z$ and $\hat{s}_z$ contribute to the scalar product, so that the final result for the correction is simply given by $\kappa m_l m_s$. To summarize, to first-order in the perturbation expansion, the energy eigenvalues in the case of the Paschen–Bach effect are given by

$$E_{nl}^{(0)} + E_{nlm_l m_s}^{(1)} = E_{nl}^{(0)} + (m_l + 2m_s) \Lambda + \kappa m_l m_s, \quad (11.76)$$

where $\Lambda = \mu_B B_{\text{ext}}$ is sometimes called the Lorentz parameter. In Fig. 11.8 we sketch a diagram of the s and p energy levels of a hydrogen atom with emphasis on the magnetic and spin–orbit corrections to the unperturbed energies (see Prob. 11.17).

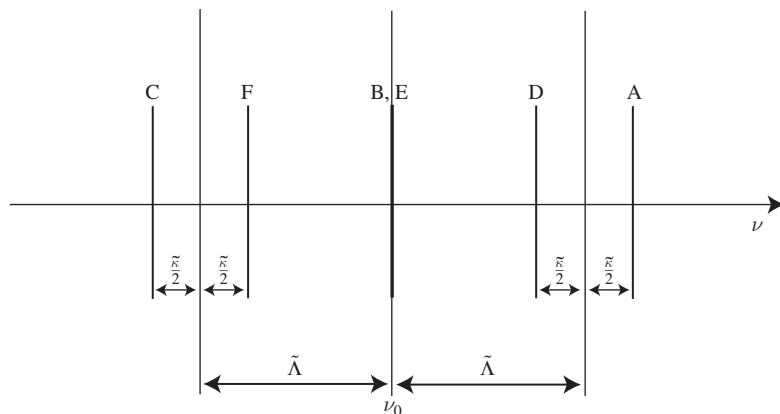


Figure 11.9

Schematic diagram of the spectroscopic lines resulting from the allowed transitions A, B, C, D, E, F of Fig. 11.8. Here ν_0 is the central frequency, i.e. the frequency of the unperturbed $s \leftrightarrow p$ transition, $\tilde{\Lambda} = \Lambda/h$, and $\tilde{\kappa} = \kappa/h$.

It is also evident from Fig. 11.8 that the $s \leftrightarrow p$ transition is profoundly affected by the magnetic and spin-orbit perturbations, even though not all the – virtually – possible transitions are actually allowed. In the electric-dipole approximation,¹² one may establish some rules that limit the permitted transitions. These are called *selection rules*, and in the present case are

$$\Delta l = \pm 1, \quad \Delta m_l = 0, \pm 1, \quad \Delta m_s = 0. \quad (11.77)$$

The first selection rule implies that transitions within the levels s and p are not allowed, whereas the second states that a photon emission or absorption may change the magnetic quantum number by at most 1. The third rule is justified by the fact that a photon exchange cannot invert the spin. In Fig. 11.9 we show the spectroscopic lines that stem from the allowed transitions depicted by vertical arrows in Fig. 11.8 (see Prob. 11.18).

11.3.3 Zeeman effect

Let us now go back to Eq. (11.68) and consider the case in which the magnetic field is weak, i.e. the second (magnetic field) term is much smaller than the third (spin-orbit) one.¹³ In this case, the “good” quantum numbers are represented by $\{n, l, j, m_j\}$, and the total Hamiltonian, according to the Dirac Picture, may be separated into

$$\hat{H} = \hat{H}_0 + \hat{H}_I, \quad (11.78)$$

¹² For the classical treatment see [Jackson 1962, Chs. 4–5]. See also Sec. 13.7 for the general quantum treatment of the interaction between atom and electromagnetic field.

¹³ For the history of the anomalous Zeeman effect see [Mehra/Rechenberg 1982–2001, I, 445–85].

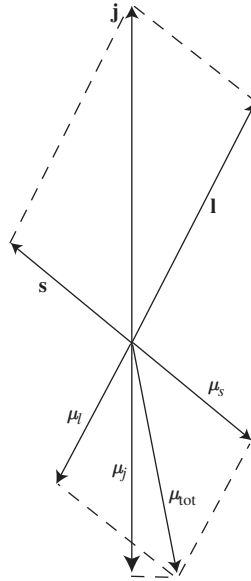


Figure 11.10

Semiclassical Landé vectorial model for the Zeeman effect. The orbital magnetic momentum μ_l (see Eq. (6.166)) is opposite to $\mathbf{l}$, whereas the spin magnetic momentum μ_s (see Eq. (6.169)) is opposite to $\mathbf{s}$. The total magnetic momentum $\mu_{\text{tot}} = \mu_l + \mu_s$, due to the different gyromagnetic ratio (see p. 225) in the two cases, is not parallel to $\mathbf{j}$. The relevant quantity for our purposes is then represented by μ_j , i.e. the projection of μ_{tot} along the direction of $\mathbf{j}$.

where

$$\hat{H}_0 = \hat{H}_A - f(r)\hat{\mathbf{l}} \cdot \hat{\mathbf{s}}, \quad (11.79)$$

and

$$\hat{H}_I = -\frac{e\hbar}{2m_e c} B_{\text{ext}} (\hat{l}_z + 2\hat{s}_z). \quad (11.80)$$

In order to find the energy shift due to the magnetic perturbation, we have to evaluate the mean value

$$E_{nljm_j}^{(1)} = \left\langle \varphi_{nljm_j} \left| \hat{H}_I \right| \varphi_{nljm_j} \right\rangle. \quad (11.81)$$

To accomplish this task, we again resort to the Landé vectorial model (see Fig. 11.10). It turns out that the total magnetic momentum μ_{tot} , due to the different gyromagnetic ratio (see p. 225) of μ_l and μ_s , is not parallel to $\mathbf{j}$.

Since the spin–orbit interaction is much larger than the magnetic one, the time scale of the “magnetic” dynamics is much larger than the corresponding spin–orbit interaction time scale. As a consequence, due to time averaging, the contribution to Eq. (11.81) of the component of μ_{tot} orthogonal to $\mathbf{j}$ vanishes. The relevant quantity for our purposes is then represented by μ_j , i.e. the projection of μ_{tot} along the direction of $\mathbf{j}$. In order to calculate μ_j , we start from

$$\mu_j = \mu_l \cos \theta + \mu_s \cos \phi = \frac{e\hbar}{2m_e c} (l \cos \theta + 2s \cos \phi), \quad (11.82)$$

where θ is the angle between $\mathbf{l}$ and $\mathbf{j}$, and ϕ is the angle between $\mathbf{s}$ and $\mathbf{j}$. Making use of the Carnot theorem in the form

$$s^2 = l^2 + j^2 - 2lj \cos \theta, \quad (11.83)$$

and

$$l^2 = s^2 + j^2 - 2sj \cos \phi, \quad (11.84)$$

we obtain

$$l \cos \theta = \frac{j^2 + l^2 - s^2}{2j} \quad \text{and} \quad s \cos \phi = \frac{j^2 + s^2 - l^2}{2j}. \quad (11.85)$$

Substituting Eqs. (11.85) into Eq. (11.82), we get

$$\mu_j = \frac{e\hbar}{2m_e c} \left(\frac{j^2 + l^2 - s^2}{2j} + 2 \frac{j^2 + s^2 - l^2}{2j} \right) = \frac{e\hbar}{2m_e c} g_L j, \quad (11.86)$$

where

$$g_L = 1 + \frac{j^2 + s^2 - l^2}{2j^2} = 1 + \frac{j(j+1) + s(s+1) - l(l+1)}{2j(j+1)} \quad (11.87)$$

is the *Landé g factor*. Such a factor should be equal to 2 from a relativistic viewpoint (and equal to 1 from a classical viewpoint). As a matter of fact, it turns out to be different from 2 (as well as from 1). This is the reason why this effect is sometimes called *anomalous Zeeman effect*.

We are now in the position to evaluate the energy shift (11.81), and write

$$E_{nljm_j}^{(1)} = \mu_B B_{\text{ext}} g_L m_j = \Lambda g_L m_j, \quad (11.88)$$

where $\Lambda = \mu_B B_{\text{ext}}$ is the already introduced Lorentz parameter. In Fig. 11.11 we draw the modifications due to the spin–orbit interaction and the Zeeman effect to the s and p levels (see Prob. 11.19), whereas in Fig. 11.12 we schematically depict the spectroscopical lines resulting from the transitions illustrated in the previous figure (see Prob. 11.20). In the Zeeman effect we have further selection rules some:

- If the spectroscopical light is observed orthogonal to $\mathbf{B}_{\text{ext}}$:

$$\Delta m_j = \pm 1 \quad (\text{for light polarized orthogonally to } \mathbf{B}_{\text{ext}}), \quad (11.89a)$$

$$\Delta m_j = 0 \quad (\text{for light polarized parallel to } \mathbf{B}_{\text{ext}}). \quad (11.89b)$$

- If the spectroscopical light is observed parallel to $\mathbf{B}_{\text{ext}}$:

$$\Delta m_j = \pm 1 \quad (\text{for light circularly polarized}), \quad (11.90a)$$

$$\Delta m_j = 0 \quad (\text{is not allowed}). \quad (11.90b)$$

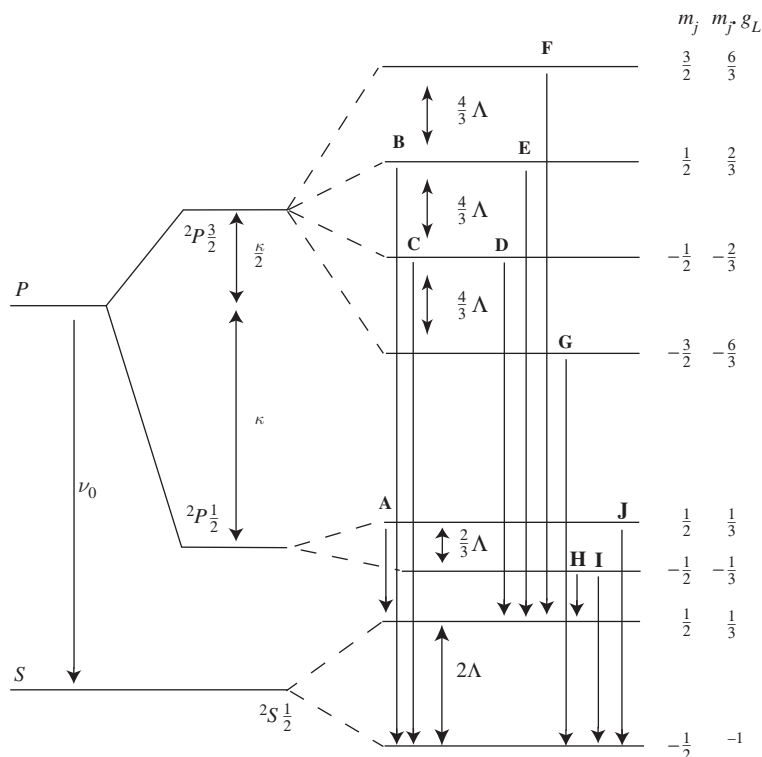


Figure 11.11

Modifications of the energy levels s and p due to the Zeeman effect. Proceeding from the left, the unperturbed levels are first affected by the spin-orbit correction, which leaves the s level unchanged. Then, each of the three resulting levels are further split due to the magnetic interaction with the external field B_{ext} . This splitting is ruled by the strength of the magnetic field (represented by Λ), the Landé g factor (which depends on l, s, j), and the eigenvalue m_j of $\hat{j}_z$. The ten possible transitions among the levels are denoted by the capital letters A, B, C, ..., J.

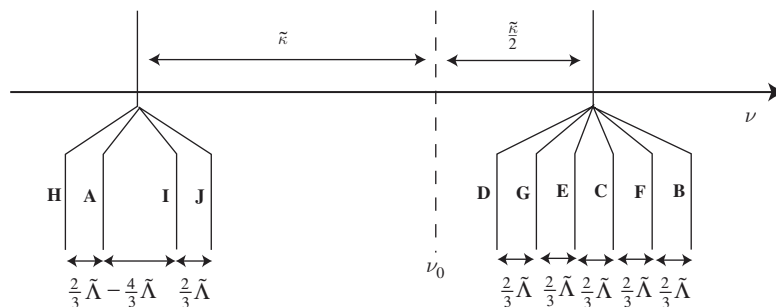


Figure 11.12

Schematic diagram of the spectroscopical lines resulting from the allowed transitions A, B, C, ..., J of Fig. 11.11. Here ν_0 is the frequency of the unperturbed $s \leftrightarrow p$ transition, $\tilde{\Lambda} = \Lambda/h$, and $\tilde{\kappa} = \kappa/h$.

It should be noted that, as the strength of the external magnetic field increases, the Lorentz parameter Λ grows linearly up to the point where two levels (the levels $^2P_{3/2,-3/2}$ and $^2P_{1/2,1/2}$ in Fig. 11.11) overlap: this intermediate situation cannot be dealt with in perturbation theory. However, for magnetic field strengths much larger than this critical value, we fall back into the framework determined by the Paschen–Bach effect.

Finally, it is interesting to emphasize that the magnetic field completely removes the degeneracy with respect to the angular momentum direction, which is opposite to what happens in the electric case – the Stark effect (see Subsec. 10.1.4).¹⁴

11.4 Relativistic corrections

In Sec. 11.2 we have neglected relativistic effects. For a full treatment of relativistic effects one should use the relativistic extension of the Schrödinger equation, i.e. the Dirac equation. However, this goes beyond the aim of this book. First-approximation relativistic corrections may be obtained by using simple heuristic arguments as shown in the present section.

Beside the spin–orbit interaction, which should also be considered to be of relativistic nature, relativistic corrections to the hydrogen atom derive from two further effects:

- Relativistically, the kinetic energy presents the form

$$\hat{T} \simeq m_e c^2 \sqrt{1 + \frac{\hat{\mathbf{p}}^2}{m_e^2 c^2}} - m_e c^2. \quad (11.91)$$

Since

$$\sqrt{1+x} \simeq 1 + \frac{1}{2}x - \frac{1}{8}x^2 + O(x^3), \quad (11.92)$$

then, for small momenta, we obtain the expansion

$$\hat{T} = \frac{\hat{\mathbf{p}}^2}{2m_e} - \frac{1}{8} \frac{\hat{\mathbf{p}}^4}{m_e^3 c^2} = \hat{T}_{nr} + \Delta\hat{T}, \quad (11.93)$$

where $\hat{T}_{nr}$ is the non-relativistic expression for the kinetic energy.

- Suppose that we wish to measure the electron position with great precision. Then, we must make use of high energy and this produces electron–positron couples. For position uncertainties,

$$\Delta x \simeq \frac{\hbar}{m_e c}, \quad (11.94)$$

¹⁴ This is due to the fact that the magnetic field is an axial vector, i.e. it changes sign upon reflection on a plane that contains its direction, while the electric field is a polar vector (see Subsec. 6.1.1).

we must take into account the creation of virtual positron–electron pairs, since such an uncertainty is of the order of the Compton wavelength (1.68), where relativistic corrections become large. A detailed computation is needed to take care of this effect. However, in the non-relativistic limit we can treat this effect as a perturbation and consider, instead of the potential energy $V(\mathbf{r})$, a perturbed potential energy $V(\mathbf{r} + \delta\mathbf{r})$, where $\delta\mathbf{r}$ is the shift due to the uncertainty (11.94). The uncertainty in the electron's position is called *Zitterbewegung*, whereby the electron does not move smoothly but instead undergoes extremely rapid small-scale fluctuations, causing the electron to see a smeared-out Coulomb potential of the nucleus. The correction will then be given by

$$\begin{aligned} V(\mathbf{r} + \delta\mathbf{r}) - V(\mathbf{r}) &= \sum_{j=\{x,y,z\}} \partial_j V(\mathbf{r}) \delta r_j + \frac{1}{2} \sum_{j,k=\{x,y,z\}} \partial_j \partial_k V(\mathbf{r}) \delta r_j \delta r_k \\ &\quad + O(\delta r^3) \\ &\simeq C \Delta V \delta r^2, \end{aligned} \quad (11.95)$$

where C is a numeric constant and the first-order terms as well as the second-order cross terms vanish in the averaging over $\delta\mathbf{r}$. From this we derive

$$\hat{H}_{\text{Darwin}} \simeq \frac{\hbar^2}{8m_e^2 c^2} \Delta V = \frac{e\hbar^2}{8m_e^2 c^2} \nabla \cdot \mathbf{E} = \frac{\pi \hbar^3 \alpha Z}{2m_e^2 c} \delta^3(\mathbf{r}), \quad (11.96)$$

where we have made use of Gauss, law

$$\nabla \cdot \mathbf{E} = 4\pi Q(\mathbf{r}), \quad (11.97)$$

where $Q(\mathbf{r})$ is the cubic charge density. For a point-like nucleus of charge Ze , we have

$$Q(\mathbf{r}) = Ze\delta^3(\mathbf{r}). \quad (11.98)$$

The Hamiltonian (11.96) is called the *Darwin term*, from the English physicist Charles Galton Darwin (1887–1962).

Since these corrections to the kinetic and potential energies are small, we can use first-order perturbation theory. In what follows we consider the joint action of both corrections, and try to estimate their relevance. First, we wish to calculate the mean value of $\hat{\mathbf{p}}^4/8m^3c^2$. Then, we have that

$$\hat{\mathbf{p}}^2 |\varphi_0\rangle = 2m_e \left(E_n^{(0)} + \frac{Ze^2}{r} \right) |\varphi_0\rangle, \quad (11.99)$$

where $|\varphi_0\rangle$ is the unperturbed eigenket of the Hamiltonian, so that

$$\begin{aligned} \langle \hat{\mathbf{p}}^4 \rangle_{\varphi_0} &= \langle \varphi_0 | \hat{\mathbf{p}}^2 \hat{\mathbf{p}}^2 | \varphi_0 \rangle = 4m_e^2 \left\langle \left(E_n^{(0)} + \frac{Ze^2}{r} \right)^2 \right\rangle_{\varphi_0} \\ &= 4m_e^2 \left[(E_n^{(0)})^2 + 2E_n^{(0)} \langle V \rangle_{\varphi_0} + \langle V^2 \rangle_{\varphi_0} \right] \end{aligned}$$

$$= 4m_e^2 \left[(E_n^{(0)})^2 + \frac{2E_n^{(0)}Ze^2}{r_0n^2} + \frac{Z^2e^4}{r_0^2n^3 \left(l + \frac{1}{2}\right)} \right]. \quad (11.100)$$

From this calculation one obtains

$$-\frac{1}{8} \frac{\langle \hat{\mathbf{p}}^4 \rangle}{m_e^3 c^2} = -\frac{(E_n^{(0)})^2}{2m_e c^2} \left(\frac{4n}{l + \frac{1}{2}} - 3 \right). \quad (11.101)$$

As for the second correction, i.e. the Darwin term (11.96), for hydrogenoid atoms we have

$$\langle \hat{H}_{\text{Darwin}} \rangle = \frac{\pi Z\alpha \hbar^3}{2m_e^2 c} |\varphi_{nlm}(0)|^2. \quad (11.102)$$

The condition $\varphi_{nlm}(0) \neq 0$ is equivalent to $l = 0$ (see Eqs. (11.55a) and (11.51)) (and therefore also $m = 0$). As a consequence, the Darwin correction will never be present together with the spin–orbit contribution. Since

$$|\varphi_{n00}(0)|^2 = \frac{1}{\pi} \left(\frac{Ze^2 m_e}{\hbar n} \right)^3, \quad (11.103)$$

then we have

$$\langle \hat{H}_{\text{Darwin}} \rangle = \frac{1}{2} m_e c^2 \left(\frac{Z\alpha}{n} \right)^4 n \delta_{l0}. \quad (11.104)$$

Now we must include the spin–orbit contribution (see Eqs. (11.65) and (11.66)) and combine it with the other two relativistic correction terms. To this purpose we must distinguish three cases:

- When $l = 0$, we only have the kinetic and Darwin corrections, and, making use of the explicit expression (11.49) for the energy eigenvalues, which for hydrogenoid atoms are

$$E_n = -\frac{Z^2 e^4 m_e}{2n^2 \hbar^2}, \quad (11.105)$$

we have

$$m_e c^2 \left(\frac{Z\alpha}{n} \right)^4 \left(\frac{3}{8} - n + \frac{n}{2} \right) = -m_e c^2 \left(\frac{Z\alpha}{n} \right)^4 \frac{n}{2} \left(\frac{1}{j + \frac{1}{2}} - \frac{3}{4n} \right), \quad (11.106)$$

because for $l = 0$, $j = 1/2$.

- When we consider the case in which $l = j - 1/2$, i.e. when the Darwin term is zero and the spin–orbit correction contributes, we obtain again

$$-m_e c^2 \left(\frac{Z\alpha}{n} \right)^4 \frac{n}{2} \left(\frac{1}{j + \frac{1}{2}} - \frac{3}{4n} \right) = -7,2 \times 10^4 \text{eV} \frac{Z^4}{n^3} \left(\frac{1}{j + \frac{1}{2}} - \frac{3}{4n} \right), \quad (11.107)$$

where we have also give the explicit expression of the sum of the contributes.

- The same result holds also for $l = j + 1/2$.

This means that we always obtain the same correction either by adding the Darwin and the kinetic corrections or by summing the spin-orbit (in both cases $l = j \pm 1/2$) and the kinetic terms. For the ground state of the hydrogen ($Z = 1, n = 1$), the correction is $\simeq 10^{-3}$ eV and though very small it is easily detectable with spectroscopy.

There are also other relativistic aspects in the hydrogen atom that – as we have said – are beyond the scope of this book; we recall that some of them are due to quantum effects such as vacuum fluctuations: the most famous being the Lamb shift.¹⁵

11.5 Helium atom

The atomic Schrödinger equation can be solved only in its simplest case: the hydrogenoid atoms. In all other cases, one has to resort to approximation methods. In this section we consider the helium atom, with two electrons and $Z = 2$. For larger Z we have ionized atoms.¹⁶ The Hamiltonian of the helium atom may be written as

$$\hat{H} = \frac{\hat{\mathbf{p}}_1^2}{2m_e} - \frac{Ze^2}{r_1} + \frac{\hat{\mathbf{p}}_2^2}{2m_e} - \frac{Ze^2}{r_2} + \frac{e^2}{r_{12}}, \quad (11.108)$$

where r_1, r_2 are the distances from the nucleus of the two electrons, and r_{12} is the distance between the electrons. We wish to study the first levels and in particular the ground state. In order to calculate the energy of the ground state, we have two methods: a perturbational approach and a variational procedure, which will be the subject of the next two subsections.

11.5.1 Perturbation method

Let us first rewrite Eq. (11.108) as

$$\hat{H} = \hat{H}_0 + \hat{H}_I, \quad (11.109)$$

where $\hat{H}_0 = \hat{H}_1 + \hat{H}_2$, and

$$\hat{H}_1 = \frac{\hat{\mathbf{p}}_1^2}{2m_e} - \frac{Ze^2}{r_1} \quad \text{and} \quad \hat{H}_2 = \frac{\hat{\mathbf{p}}_2^2}{2m_e} - \frac{Ze^2}{r_2}, \quad (11.110)$$

while $\hat{H}_I = e^2/r_{12}$. The first method consists considering the term $\hat{H}_I$ as a perturbation. Clearly, this is not a good approximation if Z is small, since, in this case, $\hat{H}_I$ is not much smaller than the two other Coulombian terms.

We consider first the case where $\hat{H}_I$ is neglected (zeroth order). In this case, the Hamiltonian is fully separable and the ground state is described by $\varphi_{100}(\mathbf{r}_1)\varphi_{100}(\mathbf{r}_2)$.

¹⁵ For a relativistic treatment of these problems see [Bjorken/Drell 1964, 52–60].

¹⁶ Examples of helioid atoms are $\text{Li}^+, \text{Be}^{++}, \dots$

As a consequence, the ground state energy will be the sum of the eigenvalues: $E_1^{(0)} = E_{100}^1 + E_{100}^2$, with (see Eq. (11.105) for $n = 1$)

$$E_{100}^1 = E_{100}^2 = -\frac{m_e c^2}{2} (Z\alpha)^2. \quad (11.111)$$

For the sake of simplicity, we move to *atomic units*,¹⁷ i.e. (see also Eq. (11.60))

$$r_0 = \frac{\hbar^2}{m_e e^2} = 1, E_0 = m_e c^2 \alpha^2 = 1, m_e = 1. \quad (11.112)$$

Therefore,

$$E_1^{(0)} = -\left(\frac{Z^2}{2} + \frac{Z^2}{2}\right) = -Z^2, \quad (11.113)$$

which for the helium atom is -4 . Then, in this approximation, the binding energy in the ground state of the helium atom is eight times the corresponding one in the hydrogen atom, i.e.

$$E_1^{(0)} = 8 \cdot 13.6 \text{ eV} \simeq -109 \text{ eV}. \quad (11.114)$$

The first excited level is degenerate because we can get it by combining $\varphi_{100}\varphi_{200}$ and $\varphi_{100}\varphi_{21m}$. As a consequence, its energy in the zeroth approximation is given by

$$E_2^{(0)} = E_{100}^1 + E_{21m}^2 = -\left(\frac{Z^2}{2} + \frac{Z^2}{8}\right) = -\frac{5}{8}Z^2 = -5 \cdot 13.6 \simeq -68 \text{ eV}. \quad (11.115)$$

Within the same approximation it is also possible to compute the limiting energy E_l , above which the spectrum becomes a mix of a continuous and a discrete part. To this aim, it is sufficient to take the limit $n \rightarrow \infty$ for one of the two electrons (say, electron 2), obtaining

$$E_l^{(0)} = E_1^1 + E_\infty^2 = -\frac{Z^2}{2} = -2 \simeq -54.3 \text{ eV}. \quad (11.116)$$

In order to get a more precise estimation of the ground-state energy, we move to the first-order perturbation analysis, and insert the repulsive Coulombian correction, i.e.

$$\begin{aligned} E_1^{(1)} &= {}_1\langle \varphi_{100} | {}_2\langle \varphi_{100} | \frac{1}{r_{12}} | \varphi_{100} \rangle_1 | \varphi_{100} \rangle_2 \\ &= \int d\mathbf{r}_1 d\mathbf{r}_2 |\varphi_{100}(\mathbf{r}_1)|^2 \frac{1}{r_{12}} |\varphi_{100}(\mathbf{r}_2)|^2, \end{aligned} \quad (11.117)$$

with (see Eq. (11.55a))

$$\varphi_{100}(\mathbf{r}) = \frac{1}{\sqrt{\pi}} Z^{\frac{3}{2}} e^{-Zr}. \quad (11.118)$$

After a direct computation (see Prob. 11.22), we finally obtain

$$E_1^{(1)} = \frac{5}{8}Z. \quad (11.119)$$

¹⁷ This corresponds to the change of variables (11.21). On a practical side, the passage to atomic units may be formally effected by choosing $e = 1, m_e = 1, \hbar = 1$. Replacing e^2 by Ze^2 , the atomic units become the so-called Coulombian units (see Prob. 11.21).

We find that, for $Z = 2$, $E_1^{(1)} \simeq -34 \text{ eV}$; this first-order perturbational result translates into a variation of the ground state from level -109 eV to level -75 eV .

11.5.2 Variational method

A second approach involves the variational method (see Sec. 10.4). In this context, one must compute the expected value of the Hamiltonian on a “smart” wave function with a special choice of parameters. At the zeroth order we have

$$\varphi_{\text{He}}(\mathbf{r}_1, \mathbf{r}_2) = \varphi_{100}(\mathbf{r}_1)\varphi_{100}(\mathbf{r}_2) = \frac{Z^3}{\pi} e^{-Z(\mathbf{r}_1 + \mathbf{r}_2)}. \quad (11.120)$$

Since one of the electrons is shielded from the proton by the other electron, we substitute Z with $\tilde{Z}$ and seek to find the $\tilde{Z}$ which minimizes $\langle \hat{H} \rangle$ (see Eq. (11.108)). Given that, for the ground state,

$$\varphi_{\text{He}}(\mathbf{r}_1, \mathbf{r}_2) = \frac{\tilde{Z}^3}{\pi} e^{-\tilde{Z}(\mathbf{r}_1 + \mathbf{r}_2)}, \quad (11.121)$$

we have (see Eqs. (11.110) and (11.116))

$$\hat{H}_1(\tilde{Z})|\varphi_{\text{He}}\rangle = \left(\frac{\hat{\mathbf{p}}_1^2}{2} - \frac{\tilde{Z}}{r_1} \right) |\varphi_{\text{He}}\rangle = \frac{\tilde{Z}^2}{2} |\varphi_{\text{He}}\rangle, \quad (11.122)$$

from which we obtain

$$\begin{aligned} \langle \hat{H}_1(\tilde{Z}) \rangle_{\varphi_{\text{He}}} &= \int d\mathbf{r}_1 \int d\mathbf{r}_2 \varphi_{\text{He}}(\mathbf{r}_1, \mathbf{r}_2) \left(\frac{\hat{\mathbf{p}}_1^2}{2} - \frac{\tilde{Z}}{r_1} \right) \varphi_{\text{He}}(\mathbf{r}_1, \mathbf{r}_2) \\ &= - \int d\mathbf{r}_1 \int d\mathbf{r}_2 \varphi_{\text{He}}(\mathbf{r}_1, \mathbf{r}_2) \frac{\tilde{Z}^2}{2} \varphi_{\text{He}}(\mathbf{r}_1, \mathbf{r}_2). \end{aligned} \quad (11.123)$$

It follows that

$$\left\langle \frac{\hat{\mathbf{p}}_1^2}{2} \right\rangle_{\varphi_{\text{He}}} = -\frac{\tilde{Z}^2}{2} + \tilde{Z} \left\langle \frac{1}{r_1} \right\rangle. \quad (11.124)$$

For the hydrogen atom we have found (see Prob. 11.13)

$$\left\langle \frac{1}{r} \right\rangle = \frac{1}{r_0} \frac{1}{n^2}, \quad (11.125)$$

with $r_0 = \hbar^2/m_e Z e^2$. In atomic units we derive

$$\left\langle \frac{1}{r} \right\rangle = \frac{Z}{n^2}, \quad (11.126)$$

and, coming back to the helium atom, for $n = 1$, we have

$$\left\langle \frac{\hat{\mathbf{p}}_1^2}{2} \right\rangle_{\varphi_{\text{He}}} = \frac{\tilde{Z}^2}{2}. \quad (11.127)$$

Table 11.1 The table of the ground-state energy of the first few helioid atoms calculated according to the perturbation and variational methods and compared with their experimental determination

Element	Z	$(E_1^{(0)})^{\text{pert}}$	$(E_1^{(0)} + E_1^{(1)})^{\text{pert}}$	E_1^{var}	E_1^{exp}
He	2	-109	-75	-77.7	-78.6
Li^+	3	-245	-194	-196.7	-197.1
Be^{++}	4	-435	-367	-369.7	-370

Therefore,

$$\left\langle \frac{\hat{\mathbf{p}}_1^2}{2} + \frac{\hat{\mathbf{p}}_2^2}{2} \right\rangle_{\varphi_{\text{He}}} = \tilde{Z}^2. \quad (11.128)$$

On the other hand, for the potential energy of the interaction-free Hamiltonian $\hat{H}_0$ (see Eqs. (11.110)), we obtain

$$\left\langle -\frac{Z}{r_1} - \frac{Z}{r_2} \right\rangle_{\varphi_{\text{He}}} = -2Z\tilde{Z}, \quad (11.129)$$

where we have again made use of Eq. (11.126) with $\tilde{Z}$ in place of Z . Finally, since (see Eq. (11.117))

$$\left\langle -\frac{1}{r_{12}} \right\rangle_{\varphi_{\text{He}}} = \frac{5}{8}\tilde{Z}, \quad (11.130)$$

we also have

$$\langle \hat{H} \rangle_{\varphi_{\text{He}}} = \tilde{Z}^2 - 2Z\tilde{Z} + \frac{5}{8}\tilde{Z}. \quad (11.131)$$

Minimizing relatively to $\tilde{Z}$, i.e. imposing

$$\frac{\partial \langle \hat{H} \rangle_{\varphi_{\text{He}}}}{\partial \tilde{Z}} = 2\tilde{Z} - 2\left(Z - \frac{5}{16}\right) = 0, \quad (11.132)$$

we finally obtain

$$\tilde{Z} = Z - \frac{5}{16}, \quad (11.133)$$

and therefore

$$\langle \hat{H} \rangle_{\varphi_{\text{He}}} = -\left(Z - \frac{5}{16}\right)^2, \quad (11.134)$$

which for the ground-state energy yields -77.5 eV. We can summarize the results of this section in Table 11.1, which shows how the accuracy of the estimation of the ground-state energy of the first few helioid atoms increases as we move from zeroth-order perturbation theory to first-order perturbation theory to end up with the variational method.

11.5.3 The spin component

It should be noted that the Hamiltonian (11.108) does not depend on the spin. However, the two electrons are identical particles (fermions) and the exchange interaction (see Sec. 7.4) comes into play. In other words, one has to multiply all the wave functions considered in this section by the spin component. For the ground state of the helium atom we shall have

$$\Psi_g(\mathbf{r}_1, \mathbf{r}_2; s) = \frac{\tilde{Z}^3}{\pi} e^{-\tilde{Z}(r_1+r_2)} \zeta(s), \quad (11.135)$$

and, since the spatial wave function is symmetric under the exchange of the two electrons, the spin wave function $\zeta(s)$ must be antisymmetric, in order for the total wave function Ψ_g of the two fermions to be antisymmetric. As a consequence, $\zeta(s)$ must be the wave function of a singlet state (see Subsec. 6.4.3) and the ground state of helium has always spin zero.

As for the wave function of the first excited level, in place of the simple product

$$\varphi_{100}(\mathbf{r}_1) \varphi_{200}(\mathbf{r}_2), \quad (11.136)$$

we shall have

$$\frac{1}{\sqrt{2}} [\varphi_{100}(\mathbf{r}_1) \varphi_{200}(\mathbf{r}_2) \pm \varphi_{100}(\mathbf{r}_2) \varphi_{200}(\mathbf{r}_1)] \zeta(s), \quad (11.137)$$

and, instead of the product

$$\varphi_{100}(\mathbf{r}_1) \varphi_{21m}(\mathbf{r}_2), \quad (11.138)$$

we must write

$$\frac{1}{2\sqrt{2}} [\varphi_{100}(\mathbf{r}_1) \varphi_{21m}(\mathbf{r}_2) \pm \varphi_{100}(\mathbf{r}_2) \varphi_{21m}(\mathbf{r}_1)] \zeta(s), \quad (11.139)$$

where this time $\zeta(s)$ can be either symmetric (triplet state together with the minus sign in the spatial part of the wave function) or antisymmetric (singlet state with the plus sign for the spatial part of the wave function). The spin-zero helium atom is called *para*-helium, whereas the spin-one is called *ortho*-helium.¹⁸

Going to the estimation of the energy of the levels, one has to evaluate expectation values of the type

$$\left\langle \Psi \left| \frac{1}{r_{12}} \right| \Psi \right\rangle, \quad (11.140)$$

which, due to the forms (11.137) and (11.139), give rise to several spin-dependent terms: This is the effect of the exchange interaction that makes the energy eigenvalues depend on the spin even though the original Hamiltonian did not.

¹⁸ This nomenclature is also used for all the two-fermion systems, such as, e.g., the electron-positron pair, called *positronium*.

11.6 Many-electron effects

The analytic solution of the Schrödinger equation is only possible in rather fortunate cases. In the one-dimensional case numerical techniques can be used with very high precision and it is relative easy to obtain results with 10 significant decimal digits. However, this is not so simple when we increase the number of dimensions. If we introduce a lattice of step a in a box of size l in D dimensions, the number $\mathcal{N}$ of points of the lattice is given by

$$\mathcal{N} = \left(\frac{l}{a}\right)^D. \quad (11.141)$$

This could still be managed in three dimensions also with non-spherically symmetric (but not too rough) potentials: 10^8 points can be used on a PC. Using our analytic command of elliptic differential equations, many efficient techniques may be found to extrapolate to the limit where $a \rightarrow 0$ and $l \rightarrow \infty$. Unfortunately, for a system with N particles we have

$$\mathcal{N} = \left(\frac{l}{a}\right)^{ND}, \quad (11.142)$$

and, as soon as $N > 2$, the number of points becomes overwhelmingly too large.

A brute force discretization method therefore fails in studying the properties of many-particle states; people in the past have developed many smart tools in order to overcome these difficulties. While in the $N \rightarrow \infty$ limit, in certain cases, thermodynamic arguments may be used to simplify the computations, a very interesting and difficult problem is provided by atoms and molecularae, where N may be rather large (but also not too large).

On the top of this problem we have to deal with the statistics. Boson statistics is usually benign, but Fermi statistics (which is the relevant one for atoms) makes everything more difficult. There are many negative effects of Fermi statistics:

- The wave function may be not positive; integrals of non-positive functions are notoriously difficult to be computed numerically.
- For the same token, the path-integral representation of the Green functions (or of the partition function) of many fermions is not positive definite and this adds extra difficulties to the already difficult method of path integrals.
- The Pauli exclusion principle implies that, roughly speaking, different fermions occupy different regions of phase space: Therefore, when the number of fermions is high, we have to explore simultaneously a quite large region of phase space, e.g. both the region of large and small momenta.

11.6.1 Variational methods

A natural approach is provided by the variational method, which, as we have seen, gives very good results for the helium atom. In the most powerful version of the method one can use the following theorem: the ground state of a certain Hamiltonian has an energy

eigenvalue that is greater than or equal to the eigenvalue of the ground state of the same Hamiltonian restricted to a subspace of the original space.

Therefore, we can consider a set of n vectors $|\psi_k\rangle$. If we are able to compute the following quantities

$$\vartheta_{j,k} = \langle \psi_j | \psi_k \rangle, \quad H_{j,k} = \langle \psi_j | \hat{H} | \psi_k \rangle, \quad (11.143)$$

the evaluation of the ground state of the Hamiltonian, restricted to the space spanned by the n vectors $|\psi_k\rangle$, involves only n -dimensional algebra and is relatively easy.

In this way, we can use as variational parameters the form of all the n vectors $|\psi_k\rangle$ and we have a quite large variational space. In some cases the quantities in Eq. (11.143) can be evaluated numerically and very good precision results are obtained.

However, the variational method needs a good starting point, and this stresses the importance of approximation methods. This problem will be studied in the rest of this section for the fermionic case, neglecting for simplicity the complications due to spin.

11.6.2 The Hartree–Fock equation

The simplest way to write a wave function of N fermions is to use the Slater determinant (see Eq. (7.20)) in terms of N orthogonal wave functions. The Hartree–Fock method consists in combining the variational approach of the previous section with the Slater determinant. We shall see how this construction may be applied to the case of Z electrons in an atom of charge Z , neglecting the spins (or equivalently assuming that they have all the same spin).

The Hamiltonian of our problem is given by

$$\hat{H} = \sum_{k=1}^Z \left(\frac{\hat{\mathbf{p}}_k^2}{2m} - \frac{Ze^2}{r_k} + \frac{1}{2} \sum_{j=1}^Z{}' \frac{e^2}{r_{j,k}} \right), \quad (11.144)$$

where the primed sum indicates that the sum has to be taken for $i \neq k$, and Z is both the charge of the nucleus and the number of electrons: we are indeed considering neutral atoms.

Our aim is to compute the expectation of $\hat{H}$ in the state $|\Psi\rangle$ formed by the Slater determinant of Z vectors $|\psi_k\rangle$. We preliminary note that if $f(x)$ is a function of one variable we have that

$$\begin{aligned} \langle \Psi | f(x_1) | \Psi \rangle &= \frac{1}{Z} \sum_{j=1}^Z \int dx \psi_j^*(x) f(x) \psi_j(x), \\ \langle \Psi | \sum_{j=1}^Z f(x_j) | \Psi \rangle &= \sum_{j=1}^Z \int dx \psi_j^*(x) f(x) \psi_j(x). \end{aligned} \quad (11.145)$$

In the case of a function of two variables $f(x, y)$ we have that

$$\begin{aligned} \langle \Psi | \sum_{j,k=1}^Z ' f(x_j, x_k) | \Psi \rangle &= \sum_{j,k=1}^Z ' \int dx \int dy f(x, y) \left[\psi_j^*(x) \psi_j(x) \psi_k^*(x) \psi_k(x) \right. \\ &\quad \left. - \psi_k^*(x) \psi_j(x) \psi_j^*(x) \psi_k(x) \right]. \end{aligned} \quad (11.146)$$

We notice that in the case of non-interacting fermions, where the total Hamiltonian is the sum of single fermion Hamiltonians, if the $|\psi_k\rangle$ is an eigenvector of the single fermion Hamiltonian with eigenvalue E_k , then the state $|\Psi\rangle$ is an eigenvector of the total Hamiltonian $\hat{H}$ with energy

$$E = \sum_{k=1}^Z E_k. \quad (11.147)$$

If we apply the previous formulae to our Hamiltonian we get

$$\begin{aligned} \langle \Psi | \hat{H} | \Psi \rangle &= \sum_{j=1}^Z \int dx \psi_j^*(x) \left(\frac{p^2}{2m} - \frac{Ze^2}{r} \right) \psi_j(x) \\ &\quad + \frac{e^2}{2} \sum_{j,k=1}^Z ' \int dx \int dy \frac{\psi_j^*(x) \psi_j(x) \psi_k^*(x) \psi_k(x) - \psi_k^*(x) \psi_j(x) \psi_j^*(x) \psi_k(x)}{|x - y|}. \end{aligned} \quad (11.148)$$

The task now is to minimize the previous expression with respect to the number Z of the functions that enter into the definition of $|\Psi\rangle$.

This is a rather complex task. Explicit equations can be written, but finding the solution of these equations numerically leads to elaborate numerical computations. A sequence of approximations can be made in order to bring the problem to a more manageable formulation and also to a physical insight of the computation.

The first approximation that, as we shall see below, is physically motivated, consists introducing an effective potential energy $V(x)$ and assuming that all the wave functions $\psi_k(x)$ satisfy the equation

$$\left(-\frac{\hbar^2}{2m} \Delta + V(x) \right) \psi_k(x) = E_k \psi_k(x), \quad (11.149)$$

with the same potential energy $V(x)$. In this way we restrict the set of variational functions, and the ground state energy that we find may be larger than the one obtained without this restriction. The advantage is that we have only one function $V(x)$ that has to be computed by minimization (instead of Z functions, as in the original case); this leads to a strong simplification. Moreover, if we consider only spherically symmetric effective potentials, the complexity of the computation significantly decreases, because, for central potential, the numerical computation of the eigenfunctions can be performed in a very effective way. In other words, the mathematical problem is reduced to that of a functional minimization with only a one-dimensional function, namely, the effective radial potential energy $V(r)$.

If we neglect the exchange term, i.e. the last term in the double integral in Eq. (11.148) (namely the one proportional to $\psi_k^*(x)\psi_j(x)\psi_j^*(x)\psi_k(x)$), we find that the potential energy $V(x)$ that minimizes the energy must satisfy the self-consistent equation

$$V(x) = -\frac{Ze^2}{r} + e^2 \int dy \frac{\rho(y)}{|x-y|}, \quad (11.150)$$

where

$$\rho(x) = \sum_{k=1}^Z |\psi_k(x)|^2. \quad (11.151)$$

Under this form the numerical computations are quite easy and can be done with high accuracy.

Similar computations can be done for molecules. However, radial symmetry cannot be used any longer. Sometimes, especially if high accuracy is not needed and qualitative results are sufficient, the computation can be done by considering the atoms as separate entities and treating the interaction among the atoms as a perturbation with the exception of the most external orbitals.

11.6.3 The Thomas–Fermi approximation

The first and simplest approximation that gives many of the characteristics of the atoms, and especially the overall dependence on Z , neglecting the oscillations of the periodic table, is the Thomas–Fermi approximation.¹⁹ It is based on the semiclassical approximation (10.124).

Let us first study what happens to Z non-interacting fermions subject to the Hamiltonian

$$\hat{H}_{tot} = \sum_{k=1}^Z \left(\frac{\hat{p}_k^2}{2m} - V(x_k) \right) = \sum_{k=1}^Z \hat{H}(p_k, x_k). \quad (11.152)$$

In order to simplify the formulae and to stress the basic principles let us present the computations in units where $\hbar = 1$, $2m = 1$ and $e = 1$. In the semiclassical approximation, we find that the energy of the ground state is given by

$$E = \int_{H < E_F} dp dx H(p, x), \quad (11.153)$$

where E_F is known as the Fermi level of energy. The density of fermions in the point x , i.e. $\rho(x)$, is given, in the same approximation, by

$$\rho(x) = \int_{H < E_F} dp. \quad (11.154)$$

The total number of fermions is Z , and this implies that

$$\int dx \rho(x) = Z. \quad (11.155)$$

¹⁹ See [Thomas 1927] [Fermi 1928].

In this case we find

$$\rho(x) = \int_{p^2 < E_F - V(x)} dp = V \Delta(x)^{3/2} \quad (11.156)$$

where $V = 4/3\pi$ (the volume of the unit sphere in three dimensions) and

$$\Delta(x) = V(x) - E_F . \quad (11.157)$$

The total energy can be written as

$$\begin{aligned} E &= V \int dx \left(\frac{3}{5} \Delta(x)^{5/2} + V(x) \Delta(x)^{3/2} \right) \\ &= \int dx \left(\frac{3}{5} V^{-2/3} \rho(x)^{5/3} + V(x) \rho(x) \right) . \end{aligned} \quad (11.158)$$

The first term is the kinetic energy and the second is the potential term.

We would like to stress the following point. If we consider the energy as a functional of ρ and we look for its stationary point, with the constraint given by Eq. (11.155), we find the equation

$$\frac{\delta E[\rho]}{\delta \rho(x)} \equiv V^{-2/3} \rho(x)^{2/3} = \lambda , \quad (11.159)$$

where λ is a Lagrange multiplier that is needed to implement the constraint given by Eq. (11.155). With the identification of E_F with λ we recover Eq. (11.156).

Therefore we can say that the energy of Z fermions can be found using the variational principle where we have to minimize the expression (11.158).

The Thomas–Fermi equation can be derived in a similar way. We obtain for the free energy functional

$$E_{TF}[\rho] = \int dx \left(\frac{3}{5} V^{-2/3} \rho(x)^{5/3} - \frac{Z\rho(x)}{r} \right) + \frac{1}{2} \int \int dx dy \frac{\rho(x)\rho(y)}{|x-y|} . \quad (11.160)$$

The variation of the previous functional leads to the equation

$$V^{-2/3} \rho(x)^{2/3} = -V(x) \equiv \frac{Z\rho(x)}{r} + \int dy \frac{\rho(y)}{|x-y|} , \quad (11.161)$$

where we have set $E_F = 0$ in such a way that the charge density $\rho(x)$ extends up to infinity. By applying the Laplacian to the previous equation, we get

$$V^{-2/3} \Delta \rho(x)^{2/3} = 4\pi (-Z\delta(x) + \rho(x)) , \quad (11.162)$$

which is the Thomas–Fermi equation.²⁰

By introducing the function

$$\omega(x) = \rho(Z^{1/3}x) , \quad (11.163)$$

the previous equation becomes independent from Z and we get

$$V^{-2/3} \Delta \omega(x)^{2/3} = -4\pi (\delta(x) + \omega(x)) . \quad (11.164)$$

²⁰ The Thomas–Fermi Equation is sometimes written for the function $\phi(x) \equiv V^{-2/3} \rho(x)^{2/3}$.

A detailed study of the previous equation shows that there is a unique solution that goes to zero at infinity as x^{-3} and satisfies the normalization condition

$$\int dx \omega(x) = 1, \quad (11.165)$$

which is equivalent to $\int dx \rho(x) = Z$.

The possibility of arriving to a Z -independent equation implies the scaling of many quantities as function of Z . In particular, the radius (R) scales as $Z^{-1/3}$ and the total binding energy (E) as $Z^{7/3}$. If we use the numerical solution of the Thomas–Fermi equation and go back to the standard units,²¹ we get

$$R = \frac{\hbar^2}{me^2 Z^{1/3}} = Z^{-1/3} 0.53 \times 10^{-11} \text{ m}, \quad E = Z^{7/3} 20.8 \text{ eV}. \quad (11.166)$$

This scaling is correct, however the numerical prefactors are typically wrong by about 30%. Indeed, the repulsive correlation among the electrons (i.e. the exchange interactions) – which is neglected here – makes the atom larger and less bound.

This is the first example of the local density functional approach where one writes more precise and accurate forms for the functional $E[\rho]$. In this way one arrives to more accurate versions of the equations (11.149), where the exchange energy is taken into account with high precision. There are many books²² and reviews on the local density functional approach (and also public-domain computer codes): the whole subject is quite active.

Summary

- First, we have shown how to apply to the hydrogen atom *Kepler's solution* for the problem of many interacting bodies, i.e. by only considering one-to-one interactions.
- The previous treatment has provided us with a *radial Schrödinger equation* that, with suitable variable changes, can be solved leading us to find the eigenvalues $E_n = -(1/n^2)E_0$ and the first total eigenfunctions.
- We have then studied the behavior of the atom in a magnetic field, by distinguishing the case in which the field is strong and the spin–orbit interaction can be treated as a small perturbation (*Paschen–Bach effect*), and the case when the external magnetic field is weak and can be treated as a perturbation (*Zeeman effect*).
- Apart from the spin–orbit interaction, two further *relativistic corrections* have been considered: the kinetic-energy term and the Darwin term. When the spin–orbit term is absent, the Darwin term is present and vice versa. However, we have shown that summation of one of the two with the kinetic-energy term gives the same result.

²¹ See [Landau/Lifshitz 1976b].

²² See [Fiolhais et al. 2003].

- We have then treated the *helium atom*, whose Schrödinger equation can be solved only by making use of approximation methods. We have in particular shown that perturbation and variational methods lead to similar results.
- Finally, we have considered *multielectron effects*, by making use of the Hartree–Fock method and of the Thomas–Fermi approximation.

Problems

- 11.1** Derive Eq. (11.6) from Eq. (11.2) and Eq. (11.8).
- 11.2** For the example of Subsec. 11.2.1, show that $\mathbf{P}$, $\mathbf{R}$ and $\mathbf{p}$, $\mathbf{r}$ are two pairs of classically conjugate variables.
- 11.3** Prove the commutators (11.11).
- 11.4** Study the behavior of the function
- $$y(x) = \frac{a}{x^2} - \frac{b}{x},$$
- with $a, b > 0$ and for $x > 0$.
- 11.5** Compute the relative deviation of the reduced mass from the electron mass in the case of the hydrogen atom.
- 11.6** Derive Eq. (11.23b).
- 11.7** Derive Eq. (11.32).
- 11.8** Prove the result (11.39).
- 11.9** Prove that the degree of degeneracy of the n -th energy eigenvalue of the hydrogen atom is equal to n^2 .
- 11.10** Using Eqs. (11.43), (11.44), and (11.48), find the first radial wave functions of the hydrogen atom for $n = 1, n = 2, n = 3$.
- 11.11** Find the correct normalization factors for the radial wave functions listed in Eqs. (11.51).
- 11.12** Find the radial probability densities corresponding to the radial wave functions listed in Eqs. (11.51).
- 11.13** Compute the mean value of $(r/r_0)^j$ for $j = 2, 1, -1, -2, -3$ for the first eigenfunctions of an electron in a hydrogen atom.
- 11.14** Check the correctness of the normalization of Eqs. (11.55a).
- 11.15** Compute the eigenfunctions φ_{210} , $\varphi_{21,\pm 1}$, φ_{310} , $\varphi_{31,\pm 1}$, φ_{320} , $\varphi_{32,\pm 1}$, and $\varphi_{32,\pm 2}$ for a hydrogenoid atom.
- 11.16** Compute the energy-level corrections due to the spin–orbit interaction for the levels s , p , d , and f in the case of the hydrogen atom. Use the results of this calculation to draw a more refined version of Fig. 6.9.
- 11.17** Compute the energy-level corrections due to the Paschen–Bach effect (included in the spin–orbit term) for the levels s and p , which are necessary to draw Fig. 11.8. (*Hint*: First determine the magnetic correction as a function of m_l and m_s . Then, find the further spin–orbit modification, again as a function of m_l and m_s .)

- 11.18** Use the selection rules (11.77) to find the six allowed transitions among the levels depicted in Fig. 11.8. Compute the frequencies of the resulting transitions depicted in Fig. 11.9.
- 11.19** Compute the energy-level corrections due to the Zeeman effect (included the spin-orbit term) for the levels s and p , which are necessary to draw Fig. 11.11. (*Hint*: First determine the spin-orbit correction (see Subsec. 11.3.1). Then, find the further magnetic modification due to the external field as a function of m_j and g_L .)
- 11.20** Use the selection rules (11.89a) to find the ten allowed transitions among the levels depicted in Fig. 11.12. Compute the frequencies of the resulting transitions depicted in Fig. 11.11.
- 11.21** From the definition of atomic units (11.112) deduce the corresponding units for the fundamental physical quantities of length, mass, time, and electronic charge.
- 11.22** Derive the result (11.119).

Further reading

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- Fiolhais, C. Nogueira, F., and Marques, M. (eds.), *A Primer in Density Functional Theory*, Berlin: Springer, 2003.
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